GPF-Net: Graph-Polarized Fusion Network for Hyperspectral Image Classification

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***Abstract*— Recently, there has been growing interest in hyperspectral images (HSIs) classification tasks, with both graph neural networks (GNN) and convolutional neural networks (CNNs) proving to be effective means of analysis. GNN can better capture the spatial structure of HSIs in large target irregular regions through superpixel segmentation, while CNN can refine classification tasks by processing pixel-level features in small target regular regions. However, neither GNN nor CNN models alone can simultaneously consider superpixel-level and pixel-level features to cover both large and small target regions. To fully utilize the strengths of GNN and CNN, we propose a novel model called the graph-polarized fusion (GPF) network. The GPF consists of two branches: the fused GNN (FGNN) classifier in the GNN branch conducts feature learning on large, irregular target regions using both graph convolutional network (GCN) and graph attention network (GAT) as feature extraction operators. The features are integrated using three aggregators, namely, min, max, and weighted add, followed by updating the nodes through 2-D convolutional layers. The polarized neural network (PNN) classifier of the CNN branch primarily works on small, target regular regions using polarized self-attention (PSA) to conduct high-resolution processing on the two dimensions of space and channel without increasing time loss. In addition, GPF employs residual connections to extract features from long distances and multiangles. It also uses weighted fusion to integrate the superpixel-level and pixel-level features obtained from the two branches. Rigorous experiments on five real datasets demonstrate that GPF can fully mine the latent features of HSIs, achieving competitive results compared with other state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), graph neural network (GNN), hyperspectral image (HSI) classification, polarized self-attention (PSA), weighted fusion.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are usually composed of multiple bands that can reveal spectral information with rich features, so they are becoming increasingly important in fields, such as environmental protection, agricultural production, and the military due to advancements in hyperspectral imaging technology [1]. However, the high-dimensional features of HSIs make some properties of these images very different from RGB images, such as sparse data distribution and offset centroid [2]. Consequently, traditional RGB image processing methods cannot be applied to HSIs effectively. Therefore, a range of image processing methods has been developed for HSIs classification, detection, and resolution enhancement. Among these fields, HSIs classification has received widespread attention [3], [4].

HSIs classification refers to determining the feature category to which each pixel belongs [2], [5]. In the early days, machine learning methods were used for HSIs classification, which usually consisted of two stages: feature extraction and classification. For example, principal component analysis (PCA) [6], linear discriminant analysis (LDA) [7], and representation multikernel learning (RMKL) [8] had been proposed to extract spectral information. In addition, to fully exploit and utilize the local pixel dependencies in HSIs, spatial–spectral combined kernel filters [9], kernel Hilbert space embedding [10], and Markov random fields [11] based on spatial information were proposed. However, the vast majority of these methods relied on manually designed features, which required a high level of expertise, and there were potential issues with the designed features.

Due to their inherent limitations, traditional machine learning methods may struggle to extract deep-level features, which can greatly affect the accuracy of classification results. In contrast, deep learning (DL) can automatically extract deep-level features [12], [13], greatly improving the accuracy. Convolutional neural network (CNN), as an integral part of DL, plays an extremely important role in the field of HSIs classification [14]. For instance, after analyzing the advantages and disadvantages of 2-D CNN and 3-D CNN, Yu et al. [15] proposed a hybrid spectral convolutional neural network (HybridSN), which achieved a classification accuracy far exceeding that of machine learning methods. The spectral–spatial residual network (SSRN) proposed by Zhong et al. [16] extracted deep-level features by stacking multiple spectral–spatial residual blocks, avoiding the

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complicated manual extraction process. Besides, with the rise of the attention mechanism, in order to better capture the features of HSIs, attention-based adaptive spectral–spatial kernel ResNet (A^2S^2KResNet) proposed by Roy et al. [17] added an attention spectrum space core based on the residual network and was trained in an end-to-end manner to improve the classification effect. Qu et al. [18] proposed the triple-attention-based parallel Network (TAP-Net), which utilizes channel, spectral, and spatial attention mechanisms to generate parallel subnetwork partition maps with consistent spatial resolution. This approach achieved state-of-the-art performance on three datasets.

Although CNN has great advantages in HSIs classification, it still has two major problems as follows. First, Dong et al. [19] pointed out that when the number of training samples is small, the classification performance of CNN decreases, and the overfitting phenomenon is obvious. However, labeling a large number of samples is an extremely resource-intensive task. Second, there are correlations between different land types in HSIs, but they may not be adjacent in the image. CNN, on the other hand, is designed for processing Euclidean data and can extract rich information from close-range regions [20]. When dealing with nonadjacent regions, CNN cannot achieve matching classification accuracy at a huge time penalty.

Graph theory has been widely applied to HSIs due to its excellent ability to handle non-Euclidean data. For example, in the field of HSIs processing, Cai et al. [21] proposed a distributed algorithm based on graph signal processing (GSP) and superpixel segmentation for hybrid noise removal of large-size HSIs, while Zhang et al. [22] proposed a spectral analysis method for HSIs based on multilayer GSP to study the fundamental features of HSIs with the advantage of GSP. Hong et al. [23] proposed learnable manifold alignment (LeMA), which learns the joint graph structure directly from the data and further captures the distribution of the data by propagation based on the graph labels. In the field of HSIs classification, graph-based models are more suitable for dealing with disjoint but related to land types than CNN. Therefore, many graph-based methods in this field have been developed. For instance, in non-DL domains, an iterative multitask regression (IMR) was proposed in [24] by learning to propagate labels on graph structure for HSI classification. Weighted sparse graph regularization (WSGR) proposed by Xue et al. [25] combined spatial information during preprocessing and postprocessing stages and introduced an adaptive modeling weight scheme based on Gaussian kernels, resulting in an excellent performance. In DL domains, the spatial–spectral graph convolutional networks (S^2GCNs) proposed by Qin et al. [26] explicitly utilized the adjacent nodes in the graph to approximate the convolution. This method made full use of the spatial information of the current pixel. However, treating each pixel in the HSIs as a node of the graph inevitably led to huge time complexity. Hong et al. [20] developed a new mini-batch GCN (miniGCN) to alleviate this problem. It allowed a large-scale GCN to be trained in a mini-batch fashion and combined with CNN to improve classification accuracy. Wan et al. [27] used a superpixel segmentation approach to construct a graph to reduce time complexity, in contrast to Hong's approach. To address the limitation of GCN in capturing only single-scale spatial information, Wan incorporated multiorder neighbor nodes into the construction of the adjacency matrix. In addition to GCN, graph attention network (GAT) [28] is another significant algorithm of graph neural network (GNN) and has also made outstanding achievements in the HSIs classification. Wang et al. proposed the spectral–spatial GAT (SSGAT) [29] as a semisupervised classification method for HSIs. SSGAT utilized a graph structure with all samples acting as nodes, and the establishment of an edge set enabled the full utilization of spectral–spatial information to improve classification accuracy. Spectral pyramid GAT proposed by Wang et al. [29] used 3-D CNN to extract multiscale features in the spectral dimension while connecting GAT to explicitly extract high-order spatial information, and the classification accuracy was significantly improved.

GCN and GAT are two major algorithms of GNN, and both have made outstanding contributions to the classification of HSIs. In essence, they belong to two different feature extraction operators. GCN utilizes the graph Laplacian matrix, whereas GAT utilizes the attention mechanism to dynamically change the weight between nodes when extracting features. If they are combined to extract features from different angles for HSIs classification, can they produce better effects? At the same time, GNN and CNN have their own advantages when dealing with HSIs. By superpixel segmentation, pixels that are not originally adjacent but are related to acquire a physical sense of adjacency, and the GNN can handle long-range irregular regions in large targets through this adjacency. However, a weakness of GNN is its inability to refine small target regions, whereas CNN can compensate for this drawback. CNN uses multiple convolutional kernels to extract features of small, regular target regions and achieve fine classification at a short range. Two branches can be set to process HSIs from the perspectives of large and small targets, and their outputs can be fused to solve the problem of a single model being unable to consider both long and short ranges.

This article is based on two branches of GNN and CNN. For the GNN branch, we formulate a novel classifier called fused GNN (FGNN), while in the CNN branch, we adopt the polarized neural network (PNN) classifier. This approach allows us to effectively integrate the strengths of both GNN and CNN for improving the classification performance. Polarized here refers to the polarized self-attention (PSA) mechanism, which has obvious advantages in pixel-level classification tasks. Specifically, superpixel segmentation is applied to the FGNN to transform pixels into superpixels, and the adjacency relationship between superpixels is utilized to create the graph. Feature extraction operators include GCN and GAT, and then, the extracted features are aggregated by min, max, and weighted add aggregators. Finally, node updates are achieved through the application of 2-D convolutional layers. In the PNN architecture, a band selector is utilized to eliminate redundant bands, and the resulting filtered pixel-level features are sent to the point-depth convolution layer after undergoing processing by the PSA blocks. In order to capture long-distance and

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multiangle features, the PNN incorporates residual connections in its architecture. In addition, a weighted fusion of the outputs produced by the two branches is performed at the end of the model, followed by a fully connected layer that adjusts the dimensionality of the output. The classification label Y is obtained by softmax. The main contributions of this article are as follows.

1) After analyzing the characteristics of GCN and GAT, we propose a novel FGNN. This classifier employs three aggregators to effectively aggregate the extracted features and fully utilize the spatial–spectral information in HSIs.
2) To extract the spatial–spectral information in HSIs more effectively, we utilize a band selector to remove redundant bands and then introduce the PSA mechanism to design the PNN classifier of the CNN branch. Moreover, we incorporate residual connections to capture long-distance information and multiangle features.
3) This article proposes a network architecture applied to HSIs classification. It utilizes a weighted fusion method to combine the outputs of GNN and CNN branches, enabling the model to capture both small target regular regions and large target irregular regions in HSIs.
4) We conduct extensive experiments on five famous datasets (Indian Pines, University of Pavia (PaviaU), Salinas, WHU_Hi_LongKou, and Botswana), and the results show that our proposed graph-polarized fusion (GPF) outperforms the state-of-the-art HSIs classification models.

The rest of this article is organized as follows. Section II presents the preliminaries, Section III details our proposed method, and Section IV presents experimental results and analysis. Finally, Section V concludes this article and presents priorities for future work.

II. Preliminaries

A. Graph Convolutional Network

The semisupervised GCN proposed by Kipf and Welling [30] had been proven to be an effective tool for dealing with non-Euclidean data. In order to reduce time complexity, he pointed out that Chebyshev polynomials can be used to approximate the convolution kernels of a graph. In detail, a convolutional filter g_θ parameterized by θ is represented. Chebyshev polynomials are used to approximate $P_{\hat{k}}(x)$ up to order $\hat{K}$. This allows graph convolution to be expressed as follows:

$$g_\theta \star x \approx \sum_{\hat{k}=0}^{\hat{K}} \Theta_{\hat{k}} P_{\hat{k}}(\hat{L})x \tag{1}$$

where x here means the graph signal and $\Theta_{\hat{k}}$ denotes the kth coefficient of the Chebyshev polynomial. $\hat{L}$ is obtained from the normalized Laplacian matrix L of the graph through a series of operations. In detail, $L = I - D^{-(1/2)}AD^{-(1/2)} = U\Lambda U$ and $\hat{L} = (2/\lambda_{\max})L - I$. Among them, I is the identity matrix matching the number of graph nodes and D is the degree matrix of the graph. The adjacency matrix is represented by A, and Λ is the diagonal matrix filled with the eigenvalues of L. Besides, $\lambda_{\max}$ represents the maximum eigenvalue of L, and U is the eigenvector matrix of L. For (1), to further increase its stability and practicability, the author made $\lambda_{\max}$ approximately equal to 2 and set the number of items of the Chebyshev polynomial to 2, that is, $\hat{K} = 1$. Then, the calculation can be expressed as follows:

$$g_\theta \star x \approx \Theta'_0 x + \Theta'_1(L - I)x = \Theta'_0 x - \Theta'_1 D^{-\frac{1}{2}} A D^{-\frac{1}{2}} x \tag{2}$$

where Θ'_0 and Θ'_1 are two free parameters shared by the whole graph. The practice has proved that an excessive number of parameters may lead to overfitting problems, so further limiting the number of parameters may be beneficial. Therefore, the author restricted $\Theta'_0 = -\Theta'_1$ and obtained the following expression:

$$g_\theta \star x \approx \Theta(I + D^{-\frac{1}{2}} A D^{-\frac{1}{2}})x. \tag{3}$$

In order to eliminate the numerical instability and gradient explosion problems that may be caused by (3), the following renormalization techniques are introduced: $I + D^{-(1/2)}AD^{-(1/2)} \rightarrow \tilde{D}^{-(1/2)}\tilde{A}\tilde{D}^{-(1/2)}$, with $\tilde{A} = A + I$ and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Finally, (3) is written as follows:

$$G_\tau = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X \Theta \tag{4}$$

where $X \in \mathbb{R}^{N\times C}$ is the graph signal and $\Theta \in \mathbb{R}^{C\times F}$ is now a matrix of filter parameters. $G_\tau \in \mathbb{R}^{N\times F}$ represents the output of graph convolution.

According to (4), the node information obtained by GCN can be propagated between labeled samples and unlabeled samples to obtain discriminative features. However, due to the Laplacian matrix, if multiple layers of GCN are superimposed, it will lead to oversmoothing. In addition, a shallow GCN cannot extract the features well in the graph. Therefore, we design a residual GCN (Res_GCN) to make the extracted features more representative. Res_GCN adds a residual connection between GCN layers, which can perform better feature extraction and improve classification results. The details will be given in Section III.

B. Graph Attention Network

GAT proposed by Veličković et al. [28] is obtained on the basis of GCN. Different from GCN, GAT follows the self-attention strategy, that is, it computes the hidden representation of each graph node by assigning attention to its neighbor nodes.

For the single-layer *graph attentional layer* in the GAT architecture, its input is a set of node features $N = [n_1, n_2, \ldots, n_Z]$, where Z refers to the number of nodes. In order to enhance the expressiveness of the model, we apply the weight matrix $\hat{W} \in \mathbb{R}^{F'\times F}$ to linearly transform N, where F' indicates the number of output features for each node and F indicates the number of input features for each node. A shared attention mechanism a is then utilized to perform self-attention on every node, and the attention coefficient is calculated as follows:

$$e_{ij} = a(\hat{W}_{n_i}, \hat{W}_{n_j}). \tag{5}$$

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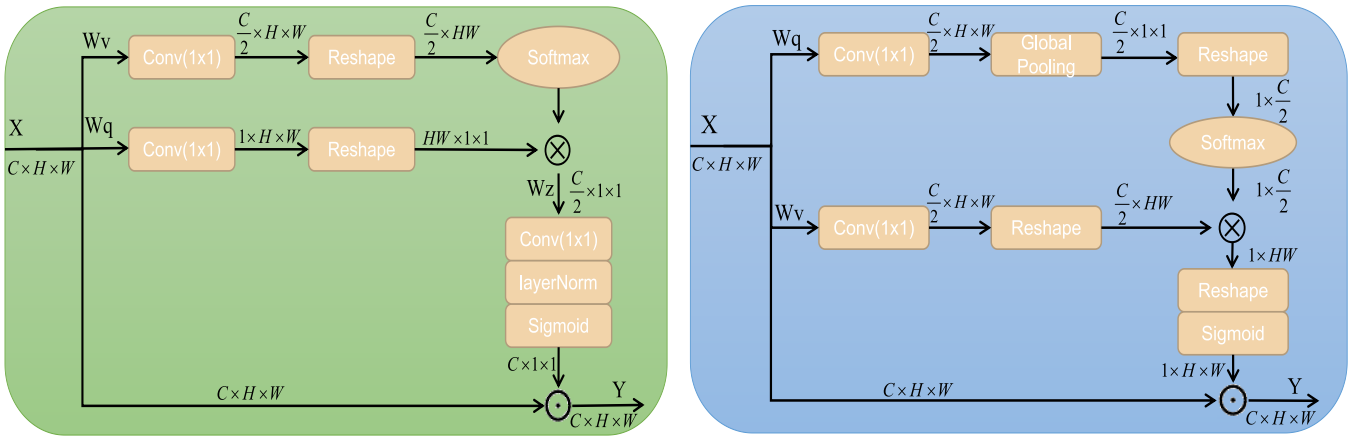


Fig. 1. Overall structure of the COA and SOA modules, where $\otimes$ is the matrix multiplication operator and $\odot$ is the Hadamard product operator. (a) COA module. (b) SOA module.

The coefficient e_{ij} represents the importance of node j's features relative to node i. Then, we normalize all the correlation coefficients of j based on the softmax, so that the coefficients are easy to compare across different nodes

$$\alpha_{ij} = \text{Softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k\in N_i}\exp(e_{ij})} \tag{6}$$

where N_i denotes the neighborhood of node i. The graph attentional layer is a single-layer feedforward neural network that uses LeakyReLU (with negative input slope of 0.2) for activation. Fully expanded out, the coefficients computed by the attention mechanism can be expressed as follows:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^{\text{T}}[\hat{W}_{n_i}||\hat{W}_{n_j}]))}{\sum_{k\in N_i}\exp\left(\text{LeakyReLU}(a^{\text{T}}[\hat{W}_{n_i}||\hat{W}_{n_k}])\right)} \tag{7}$$

where a^{T} represents the transpose of a, which is the weight vector. Besides, $||$ is the concatenation operation, and the first layer adopts a multihead attention mechanism. The final output features for each node are calculated as follows, where $\sigma(.)$ denotes a nonlinear function:

$$n_i' = \|_{k=1}^{K}\sigma\left(\sum_{j\in N_i}\alpha_{ij}^k\hat{W}^k n_j\right). \tag{8}$$

The GAT mentioned in this article is the same as that in [28]. To more abundantly extract HSIs information, K is set to 8.

C. Polarized Self-Attention

PSA [31] is a new channel-spatial attention mechanism. This approach maintains high resolution in both the spatial and channel dimensions, allowing for the acquisition of comprehensive features from these dimensions. The HSIs classification is essentially a pixel-level classification problem, which requires more time and computational resources compared with ordinary image classification tasks. For tasks involving pixel-level classification, pure 2-D CNN networks are typically compressed into two parts, channel and spatial, to reduce computational costs. However, pixels with the same semantic features are often highly correlated, so compressing the channel and spatial components can lead to a loss in final classification accuracy. Therefore, researchers typically attempt to design more complex network architectures for such tasks to handle the interpixel correlations and improve classification accuracy [32], [33].

TABLE I
COMPARISON RESULTS OF PSA BLOCK AND OTHER ATTENTION BLOCKS IN THREE ASPECTS

Method	Channel Resolution	Spatial Resolution	Non-linearity	Complexity
NL [34]	C	$[W, H]$	SM	$C^2WH + CW^2H^2$
GC [35]	$C/4$	–	$SM + ReLU$	CWH
SE [36]	$C/4$	–	$ReLU + SD$	CWH
CBAM [37]	$C/16$	$[W, H]$	SD	CWH
DAN [38]	$C/8$	$[W, H]$	SM	$C^2WH + CW^2H^2$
PSA [39]	$C/2$	$[W, H]$	$SM + SD$	CWH

The comparison between the PSA block and other attention blocks in terms of resolution, nonlinearity, and time complexity is shown in Table I, where SM refers to softmax, while SD stands for sigmoid. The PSA block is divided into two modules: channel-only attention (COA) and spatial-only attention (SOA). Fig. 1 shows the overall framework of the two modules.

We denote the feature tensor input into the PSA block as $X \in \mathbb{R}^{C\times H\times W}$, where C, H, and W are the numbers of channel, height, and width of X, respectively. $X = \{x_i\}_{i=1}^{HW}$, where $x_i \in \mathbb{R}^C$ is a feature vector along the channel dimension. PSA block Att(.) takes X as input and produces tensor Y as output, where $Y \in \mathbb{R}^{C\times H\times W}$.

1) Channel-Only Attention: The COA module deals with the channel dimension, and the calculation process is as follows:

$$\text{Att}^{\text{ch}}(X) = F_{\text{SG}}[\hat{W}_z((\gamma_1(\hat{W}_v(X)) \times F_{\text{SM}}(\gamma_2(\hat{W}_q(X)))))] \tag{9}$$

where $\text{Att}^{\text{ch}}(.)$ represents the channel processing module in the PSA block; and $\hat{W}_z$, $\hat{W}_v$, and $\hat{W}_q$ represent weight matrices of 1×1 convolutional layers, respectively. γ_1 and γ_2 are two tensor reshape operators. Besides, $F_{\text{SM}}(.)$ is a softmax

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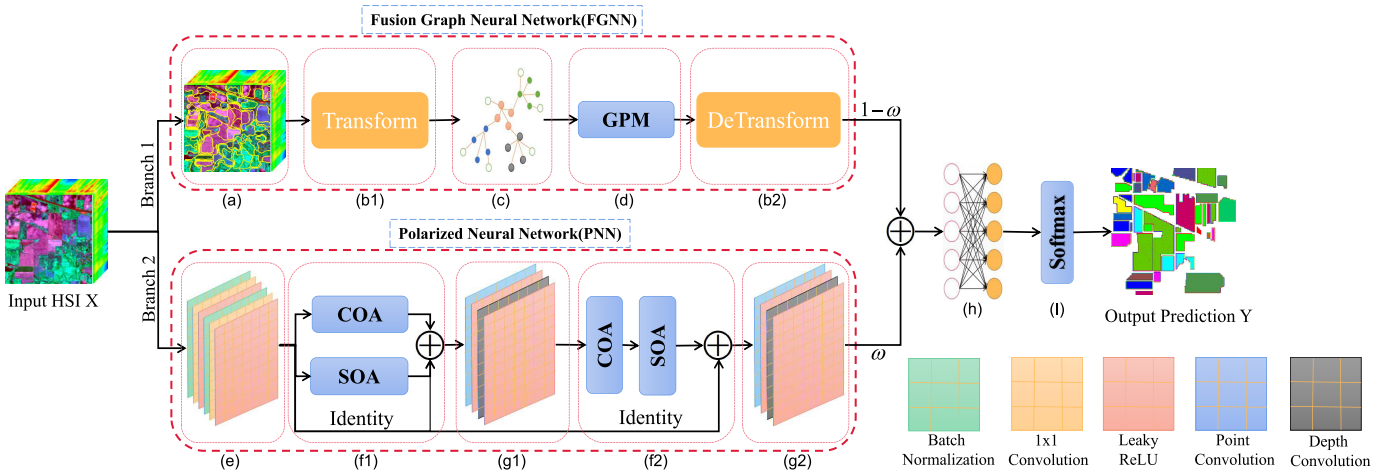


Fig. 2. Overall structure of the GPF, including nine parts. (a) LDA_SLIC module. (b1) and (b2) Image and graph interconversion module. (c) Graph presentation. (d) GPM. (e) Band selection module. (f1) and (f2) PSA block. (g1) and (g2) Point-depth convolution module. (h) Fully connected layer. (I) Softmax.

operator, while $F_{\mathrm{SG}}(.)$ is a sigmoid operator, and $\times$ is the matrix dot-product operation. The number of internal channels input to $\hat{W}_z$, $\hat{W}_v$, and $\hat{W}_q$ is $C/2$. The output of the COA module is $\hat{Z}^{\mathrm{ch}}(X) = \mathrm{Att}^{\mathrm{ch}}(X) \odot X \in \mathbb{R}^{C\times H\times W}$, where $\odot$ essentially represents the Hadamard product operator of two matrices.

2) Spatial-Only Attention: The SOA module deals with the spatial dimension, and the calculation process is as follows:

$$\mathrm{Att}^{\mathrm{sp}}(X) = F_{\mathrm{SG}}[\gamma_3(F_{\mathrm{SM}}(\gamma_1(F_{\mathrm{GP}}(\hat{W}_q(X)))) \times \gamma_2(\hat{W}_v(X)))] \tag{10}$$

where $\mathrm{Att}^{\mathrm{sp}}(.)$ represents the spatial processing module in the PSA block, and $\hat{W}_v$ and $\hat{W}_q$ represent the weight matrices of standard 1×1 convolutional layers, respectively. γ_1, γ_2, and γ_3 are three tensor reshape operators. $F_{\mathrm{SM}}(.)$, $F_{\mathrm{SG}}(.)$, and $\times$ are consistent with those in the COA module, but F_{GP} is a global pooling operator. The number of internal channels input to $\hat{W}_q$ and $\hat{W}_v$ is $C/2$. The output of the SOA module is $\hat{Z}^{\mathrm{sp}}(X) = \mathrm{Att}^{\mathrm{sp}}(X) \odot X \in \mathbb{R}^{C\times H\times W}$.

3) Composition: There are two combinations of output results of the COA and SOA modules, sequential and parallel. For the sequential layout, the output of the PSA block is as follows:

$$\begin{aligned}\mathrm{PSA}_s(X) &= \hat{Z}^{\mathrm{ch}}(\hat{Z}^{\mathrm{sp}}) \\ &= \mathrm{Att}^{\mathrm{ch}}(\mathrm{Att}^{\mathrm{sp}}(X)\odot X)\odot \mathrm{Att}^{\mathrm{sp}}(X)\odot X\end{aligned} \tag{11}$$

while the output of the PSA block placed in the parallel layout is

$$\begin{aligned}\mathrm{PSA}_p(X) &= \hat{Z}^{\mathrm{ch}} + \hat{Z}^{\mathrm{sp}} \\ &= \mathrm{Att}^{\mathrm{ch}}(X)\odot X + \mathrm{Att}^{\mathrm{sp}}(X)\odot X.\end{aligned} \tag{12}$$

In this article, the PSA blocks are embedded in two locations, but the arrangement of COA and SOA in these locations differs. Four different arrangements are experimented with in Section IV, and it was found that the arrangement of parallel first and then series yields the best results.

III. Proposed Method

As illustrated in Fig. 2, it is the overall framework of the proposed GPF for HSIs classification. We denote the input of the network as $X \in \mathbb{R}^{C\times H\times W}$, where H, W, and C represent the height, width, and number of the bands, respectively. The model is mainly divided into two branches, a total of nine modules. The processing object of the PNN is the pixel-level features, which are aimed at the small target regular regions. Band selection occurs at the beginning of this branch. The backbone of the PNN is the point-depth convolution layer embedded in the PSA blocks and joined with residual connections to enhance the feature extraction capability of the model. The details of the two PSA blocks are different, specifically reflected in the arrangement of the SOA and COA modules. The FGNN is specifically targeted at dealing with large irregular regions, and to minimize errors, it replaces the band selection operation with superpixel division, followed by constructing a graph based on adjacency. This branch also utilizes two different types of feature extraction operators for superpixels and employs three aggregators that are designed to perform efficient aggregation of the extracted superpixel-level features. The node update operation is done by 2-D convolutional layers. The weighted and fused outputs of the two branches are subjected to dimensionality adjustment through a fully connected layer. Finally, the label Y of each pixel is obtained through softmax.

A. Superpixel Division and Data Transformation

In FGNN, superpixels are employed to build the graph to computational time. Specifically, LDA [7] is first utilized to reduce the dimensionality of HSIs and remove redundant information. Next, simple linear iterative clustering (SLIC) [40] is adopted to divide HSIs into many superpixels with visual significance. These operations reduce the number of nodes, enhance the model's scalability, and enable the input of HSIs into GNN.

Since the number correspondences between superpixels and pixels are different from each other, the above segmentation

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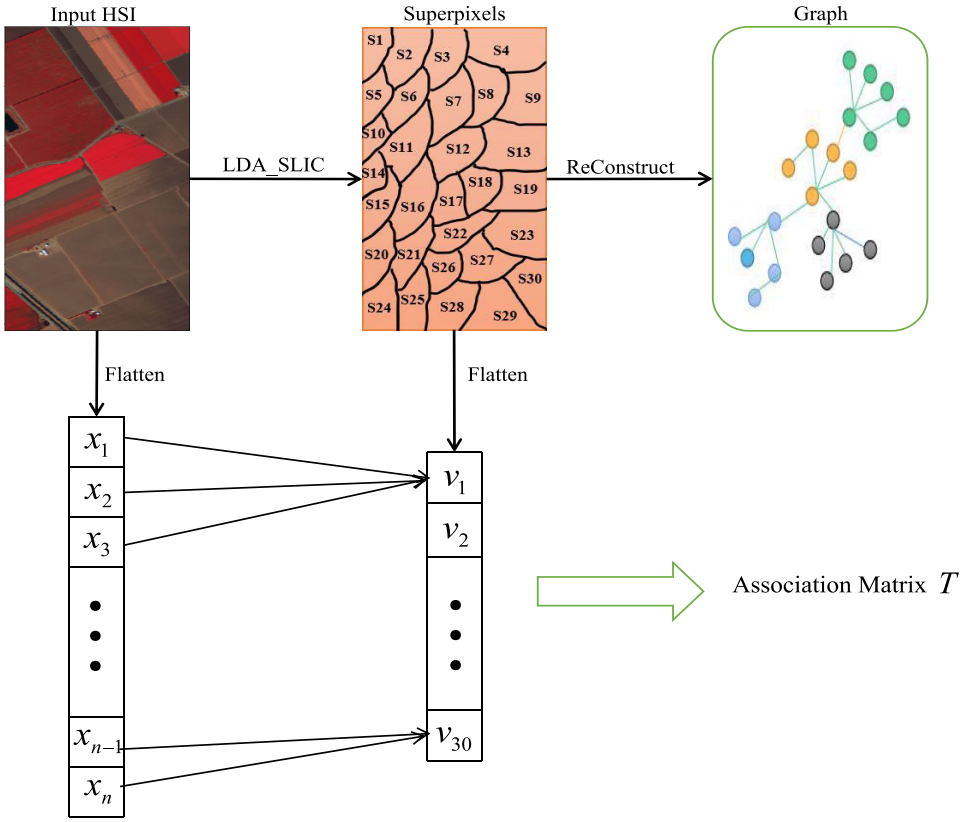


Fig. 3. Data transformation module, where x_i denotes the ith pixel in the flattened HSI and v_j denotes the average features of the pixels contained in superpixels S_j.

methods cannot be directly integrated into our network. In order to solve this problem, inspired by Jampani [41], we develop an association matrix $T \in \mathbb{R}^{\mathrm{HW}\times O}$ to realize the data transformation between superpixels and pixels, where O denotes the number of the superpixels. The details are shown in Fig. 3. Through the association matrix T, we can get

$$T_{i,j} = \begin{cases} 1, & \widetilde{X}_i \in S_j \\ 0, & \widetilde{X}_i \notin S_j \end{cases} \quad \widetilde{X} = \mathrm{Flatten}(X) \tag{13}$$

where Flatten(.) denotes the HSIs data flattened according to the spatial dimension and $T_{i,j}$ denotes the value of T in the location of (i, j). Besides, $\widetilde{X}_i$ denotes the ith pixel in $\widetilde{X}$ and S_j is the jth superpixel. Therefore, the data transformation can be expressed as follows:

$$V = \mathrm{Transform}(X; T) = \hat{T}^T \mathrm{Flatten}(X) \tag{14}$$

$$\overline{X} = \mathrm{DeTransform}(V; T) = \mathrm{Reshape}(TV) \tag{15}$$

where $\hat{T}^T$ denotes transpose of the normalized T by column and V is the set of superpixels. Transform(.; T) is a graph encoder that encodes HSIs into nodes of a graph based on T, where $\hat{T}^T$ has a dimension of $O \times$ HW and the flattened X dimension is HW $\times$ C; the two dimensions are calculated to obtain $V \in \mathbb{R}^{O\times C}$. In addition, the operation Reshape(.) is used to restore the spatial dimensions of the flattened data. Specifically, $\overline{X}$ represents the node representation after DeTransform(.;T), where $T \in \mathbb{R}^{\mathrm{HW}\times O}$ and $V \in \mathbb{R}^{O\times C}$ undergo matrix operations to obtain a column vector of dimension HW $\times$ C. Finally, Reshape(.) restores the representation to its original shape of dimension $C \times H \times W$. After the above operations, HSIs are coded as nodes in the graph and its inverse transformation, which is conducive to the feature extraction of FGNN and the feature fusion with the PNN.

B. Graph Processing Module

After the stage of superpixel division and data transformation, HSIs can be converted into non-Euclidean data and processed by GNN. Subsequently, the graph processing module (GPM) in FGNN will be introduced, which serves as the feature processing module. Fig. 4 shows its overall architecture. After encoding the pixel's latent representation of HSIs into the superpixels, the adjacency matrix is further derived based on the adjacency A, and the calculation method is as follows:

$$A_{i,j} = \begin{cases} e^{-\beta\|X^i - X^j\|_2^2}, & \text{if } X^i \text{ adj } X^j \\ 0, & \text{Otherwise} \end{cases} \tag{16}$$

where X^i denotes the ith superpixel, β denotes a steepness hyperparameter affecting the weights, and adj denotes the adjacency relationship. Then, two operators for feature extraction are utilized. Due to the difficulty of extracting deep features from the graph with a single-layer GCN and the potential oversmoothing issue caused by simply stacking GCN layers, we utilized Res_GCN to achieve the best compromise. Its architecture is shown in Fig. 5, where BN is the BatchNormalization operation and identity denotes the identity map.

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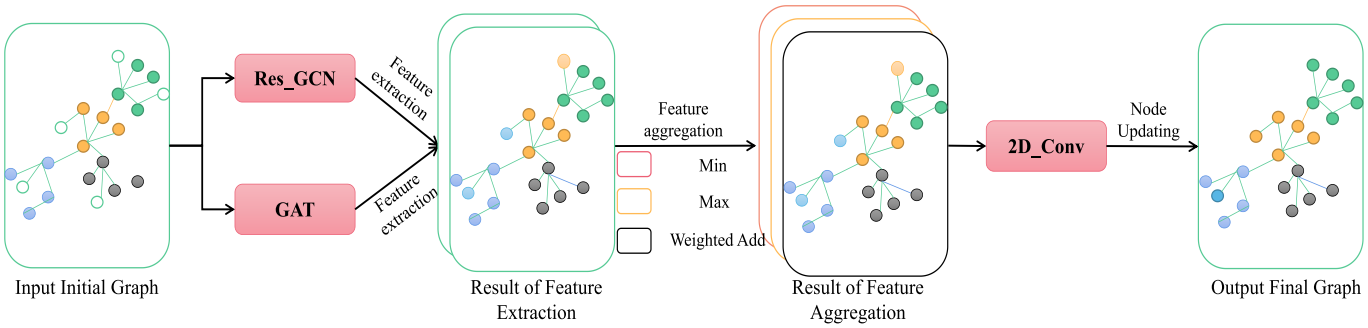


Fig. 4. Overall architecture of the GPM module.

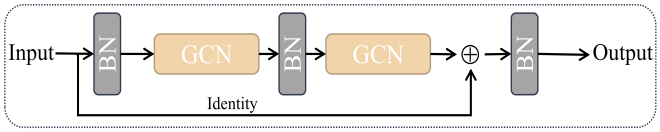


Fig. 5. Overall architecture of the Res_GCN module.

Suppose the input of GPM is an undirected graph, denoted as $G = (X, E)$, where X and E represent the nodes and edges of the graph, respectively. After extracting features through the Res_GCN operator, the output node features are expressed as follows:

$$G_{\text{Res_GCN}} = \phi(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\phi(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}\phi(X)\Theta_1)\Theta_2 + \ddot{X}). \tag{17}$$

Among them, $\tilde{D}$ and $\tilde{A}$ have the same meaning as in (4). ϕ denotes the BN operation, and Θ_1 and Θ_2 are the matrices of filter parameters. $G_{\text{Res_GCN}}$ denotes the output of the Res_GCN operator. Besides, $\ddot{X}$ is the node features after identity mapping. The output node features generated by the GAT operator are expressed as follows:

$$G_{\text{GAT}} = \|_{k=1}^{K}\sigma\left(\sum \alpha^k \hat{W}^k X\right) \tag{18}$$

where G_{GAT} is the output of GAT operator.

The Res_GCN and GAT operators extract features from different perspectives, resulting in $G_{\text{Res_GCN}}$ and G_{GAT}, respectively. To enhance the aggregation of the two node features, three aggregators are employed, namely, min, max, and weighted add, as shown in Fig. 4. Specifically, the min and max aggregators are utilized to calculate the maximum and minimum values of the nodes in $G_{\text{Res_GCN}}$ and G_{GAT}, respectively, which are useful for discrete tasks. The details can be expressed as follows:

$$\begin{aligned} G_{\text{MIN}}(X^i) &= \min\left(X^i_{\text{Res_GCN}}, X^i_{\text{GAT}}\right) \\ G_{\text{MAX}}(X^i) &= \max\left(X^i_{\text{Res_GCN}}, X^i_{\text{GAT}}\right) \end{aligned} \tag{19}$$

where G_{MIN} and G_{MAX} denote the new graph node representations generated by the two aggregators min and max, respectively. The weighted add aggregator combines the two node representations in a weighted manner, as detailed below

$$G_{\text{Wei}}(X) = f(\hat{W}_1 X_{\text{Res_GCN}} + b_1 + \hat{W}_2 X_{\text{GAT}} + b_2) \tag{20}$$

where G_{Wei} denotes the new graph node representations from the weighted add aggregator and $f(.)$ is the activation function. To achieve a better aggregation of these two node features, we update the weight matrix $\hat{W}$ and bias b after training to reach the optimal values. The resulting node features are then processed by three distinct aggregators, leading to the creation of three new representations of the graph node.

In order to better fit the feature distribution, a 3-D spatial information block is generated by stacking three new graphs, denoted as $U \in \mathbb{R}^{3\times O\times Q}$, where O is the number of superpixels and Q is the number of features of each superpixel. To process the information block U, we employ 2-D convolutional layers, enabling the extraction of spatial–spectral information contained within and ultimately achieving node updating in a more effective manner. The whole process is as follows:

$$G_{\text{Out}} = \text{2D_Conv}(\phi(U)), \quad U = \vee(G_{\text{MIN}}, G_{\text{MAX}}, G_{\text{Wei}}) \tag{21}$$

where $\vee$ denotes the stacking operation and ϕ still denotes BN operation. G_{Out} is the final output of the entire GPM, which represents the new graph node features after processing by this module. After the processing of DeTransform(.; T), G_{Out} will perform feature fusion with the output of the PNN and finally realize the classification of HSIs.

C. Band Selection and Point-Depth Convolution

Band selection happens in the PNN, while the FGNN does not, because the features we use when calculating the adjacency matrix A of the graph are the average features of all bands. If band selection is performed in the FGNN, the graph does not match its adjacency matrix, resulting in a large classification error. In PNN, the input comprises Euclidean data lacking an adjacency matrix, offering the possibility of band selection.

The multiband and big data characteristics of HSIs lead to high information redundancy, which affects classification accuracy. To reduce data redundancy, it is necessary to decrease the dimensionality of HSIs. Feature extraction and band selection are the two main means of dimensionality reduction. However, in using feature extraction for dimensionality reduction, the algorithm becomes complex, resulting in a large number of calculations. Moreover, achieving the required purpose would necessitate certain transformations, subsequently altering the

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physical meaning of the original data. In contrast, band selection refers to the process of choosing a subset of bands that are crucial among all bands in HSIs. This technique can significantly reduce the data dimension of HSIs while retaining essential information, making it more targeted [42], [43].

Inspired by Lin et al. [44], the band selection module employs two 1×1 convolutional layers to remove redundant bands. Since the 1×1 convolution was proposed, its role had been widely verified. By utilizing 1×1 convolution, the depth of the network can be increased without significantly impacting performance and can achieve the objective of eliminating redundant bands, all with very minimal time costs. Suppose the data input to the network is X; after the band selection module, the output is expressed as X^*, and the detailed process is as follows:

$$X^* = \zeta(\hat{W}_2\phi(\zeta(\hat{W}_1\phi(X) + b_1)) + b_2) \tag{22}$$

where $\hat{W}_i$ and b_i, respectively, represent the ith layer convolution kernel weight matrix and bias. ϕ is the BN operation, and $\zeta(.)$ represents the LeakyReLU activation function.

The main structure of the PNN is done by point-depth convolution layers with PSA blocks. The point-depth convolution layer is based on the depthwise separable convolution in the Xception network [45]. Traditional 2D_Conv can be used to extract contextual spatial information. However, due to the large number of weight parameters of the convolution kernels, there is a risk of overfitting issues when training samples are limited. Point-depth convolution is a deviation from traditional convolution that improves the robustness of the model and effectively overcomes the aforementioned drawbacks. Point-depth convolution with one filter per input channel can be written as follows:

$$X_i^{\tau} = \zeta\left(\hat{W}_i^D \zeta\left(\hat{W}_i^P X\right) + b_i^D\right) \tag{23}$$

where X_i^{τ} is the output features of ith point-depth convolution layer, and $\hat{W}_i^P$ and $\hat{W}_i^D$ represent the weight matrices of the ith point convolution and depth convolution kernel, respectively. b_i^D is the bias of depth convolution layer. $\zeta(.)$ represents the LeakyReLU activation function.

D. PSA Block

The PSA is described in detail in Section II. This chapter mainly introduces the orchestration of the COA and SOA modules.

The COA and SOA modules deal with the channel and spatial dimension, respectively, while ensuring the high resolution of this dimension and extremely compressing the other dimension. COA and SOA can be connected in parallel or in series. However, when connecting them in series, determining which one comes first can be a tricky question. Inspired by the convolutional block attention module (CBAM) [37], the COA module is placed in the front, and the SOA module is positioned in the back when the two modules are connected in series.

In the PNN, the PSA block is applied in two locations, respectively. According to the series and parallel connection of the two modules, there are four combinations to choose from. After conducting experimental trials, a parallel connection at the first position and a series connection at the second position were found to produce better classification performance than the other three combinations. Therefore, the final arrangement is shown in Fig. 2.

The ResNet network proposed by He et al. [46] has achieved great success in various fields, and the concept of residual connection has also had a significant impact on many fields. Residual connections can guide high-level networks to learn features in low-level networks by identity mapping while reducing model complexity and suppressing overfitting. Furthermore, residual connections prevent gradient vanishing, allowing for the retention of long-range and multiangle feature information, thereby significantly improving the learning capability of the model. The output of the PSA blocks in series and parallel is as follows:

$$\begin{aligned} X^P &= \mathrm{PSA}_P(X^*) \oplus X^* \\ X^S &= \mathrm{PSA}_S(X^*) \oplus X^* \end{aligned} \tag{24}$$

where X^P and X^S represent the outputs of the parallel and serial PSA blocks, respectively. X^* represents the output features of the band selection module. $\mathrm{PSA}_P(.)$ and $\mathrm{PSA}_S(.)$ are the outputs of the two-mode PSA blocks.

E. Feature Fusion and Classification

After HSIs are processed by FGNN and PNN, two output features with different emphases are obtained, which are recorded as X^G and X^C. Since we construct the superpixels in the FGNN, the number of labels is implicitly expanded. Thus, FGNN can to some extent overcome the limitation of few labeled samples and maintain high classification accuracy. The CNN branch equipped with PSA blocks can effectively capture boundary information across multiple dimensions, locations, and scales, thereby exhibiting impressive expressive ability.

Due to the different emphases of FGNN and PNN, the feature distributions of the two are also different. The values of ω are assigned to the two branches to achieve varying degrees of scaling for improved model integration. The fusion features are as follows:

$$X^F = \omega X^C + (1 - \omega)X^G. \tag{25}$$

The obtained fusion features are passed through a fully connected layer to map the learned distributed feature representation to the sample label space. To train the network and update the parameters, the cross-entropy function is used as the loss function, which can be expressed as follows:

$$L = -\frac{1}{R}\sum_{r=1}^{R}\sum_{j=1}^{J} y_{r,j}\log(p_{r,j}) \tag{26}$$

where $y_{r,j}$ denotes the jth element of the label set Y and $p_{r,j}$ denotes the probability that pixel r belongs to the jth class. J and R refer to the total number of categories and samples of the training datasets, respectively.

IV. Experimental Results and Analysis

This section mainly introduces the datasets used in the experiments, experimental settings, comparison results, and a brief analysis of GPF.

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TABLE II
EACH CATEGORY INFORMATION OF THE INDIAN PINES DATASET

No	Class	Train	Val	Test
1	Alfalfa	1	1	44
2	Corn-notill	15	15	1398
3	Corn-mintill	9	9	812
4	Corn	3	3	231
5	Grass-pasture	5	5	473
6	Grass-trees	8	8	714
7	Grass-pasture-mowed	1	1	26
8	Hay-windrowed	5	5	468
9	Oats	1	1	18
10	Soybean-nottill	10	10	952
11	Soybean-mintill	25	25	2405
12	Soybean-clean	6	6	581
13	Wheat	3	3	199
14	Woods	13	13	1239
15	Buildings-Grass-Trees-Drives	4	4	378
16	Stone-Steel-Towers	1	1	91
	Total	110	110	10029

TABLE III
EACH CATEGORY INFORMATION OF THE PAVIAU DATASET

No	Class	Train	Val	Test
1	Asphalt	7	7	6617
2	Meadows	19	19	18611
3	Gravel	3	3	2093
4	Trees	4	4	3056
5	Painted metal sheets	2	2	1341
6	Bare Soil	6	6	5017
7	Bitumen	2	2	1326
8	Self-Blocking Bricks	4	4	3674
9	Shadows	1	1	945
	Total	48	48	42680

TABLE IV
EACH CATEGORY INFORMATION OF THE SALINAS DATASET

No	Class	Train	Val	Test
1	Brocoli_green_weeds_1	3	3	2003
2	Brocoli_green_weeds_2	4	4	3718
3	Fallow	2	2	1972
4	Fallow_rough_plow	2	2	1390
5	Fallow_smooth	3	3	2672
6	Stubble	4	4	3951
7	Celery	4	4	3571
8	Grapes_untrained	12	12	11247
9	Soil_vinyard_develop	7	7	6189
10	Corn_senesced_green_weeds	4	4	3270
11	Lettuce_romaine_4wk	2	2	1064
12	Lettuce_romaine_5wk	2	2	1923
13	Lettuce_romaine_6wk	1	1	914
14	Lettuce_romaine_7wk	2	2	1066
15	Vinyard_untrained	8	8	7252
16	Vinyard_vertical_trellis	2	2	1803
	Total	62	62	54005

A. Data Description

The performance of the GPF is evaluated using five well-known real hyperspectral datasets, i.e., the Indian Pines dataset, the PaviaU dataset, the Salinas dataset, the WHU_Hi_LongKou dataset, and the Botswana dataset.

1) Indian Pines Dataset: The Indian Pines dataset is the earliest data used for the classification of HSIs [47], which was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992. Its spatial resolution is 20 × 20 m, while the wavelength range is 0.4–2.5 μm, and it continuously captures images of ground objects in 220 consecutive bands. Due to the adverse impact of the redundancy of water-absorbing and noise bands on classification, the Indian Pines we adopt now retain 200 bands that do not include the above two types of ground objects. The final size of the Indian Pines dataset is 145 × 145 × 200. The data have a total of 21 025 pixels, but only 10 249 of them are valid object pixels, which are divided into 16 classes. The detailed division of each class and the dataset are shown in Table II.

2) PaviaU Dataset: The PaviaU data are part of the hyperspectral data of the city of Pavia, Italy, taken in 2003 by the Airborne Reflective Optics Spectrographic Imaging System (ROSIS) in Germany [48]. The spectral imager continuously images 115 bands in the wavelength range of 0.43–0.86 μm, and the spatial resolution of the resulting images is 1.3 m. Similar to Indian Pines, the 12 bands affected by noise are included in the 115 bands. After removal, the current PaviaU dataset containing 103 bands is obtained. The size of the data is 610 × 340, so it contains a total of 207 400 pixels. However, it contains a large number of background pixels, and the total number of valuable pixels is only 42 776. These pixels contain a total of nine types of ground objects. The details and division of each category of PaviaU are shown in Table III.

3) Salinas Dataset: The Salinas dataset is an HSI of the Salinas Valley in California, USA, acquired by the AVIRIS spectrometer [18]. The size of the image is 512 × 217, the number of pixels including ground objects is 54 129, and the spatial resolution is 3.7 m. The original image has a total of 224 bands. After removing the bands affected by noise, the remaining 204 bands can be used for classification. There are 16 ground object types annotated in this dataset. The experimental division of Salinas is shown in Table IV.

4) WHU_Hi_LongKou Dataset: The dataset was collected by the Headwall Nano-Hyperspec Imaging Sensor 600 Pro (DJI M600 Pro) UAV platform in Longkou Town, Hubei, China, on July 17, 2018 [49]. The dataset has an effect size of 550 × 400 pixels, of which there are 204 542 effective pixels, and there are 270 bands from 400 to 1000 nm. The spatial resolution is about 0.463 meters, and it contains a total of nine types of ground objects. The details and division of each category of WHU_Hi_LongKou are shown in Table V.

5) Botswana Dataset: The Botswana dataset is a series of data acquired by the NASA EO-1 satellite over the Okavango Delta in 2001 [50]. After removing uncalibrated and noisy bands covering water absorption features, the dataset contains a total of 145 bands, with an image size of 1476 × 256 and a

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TABLE V
EACH CATEGORY INFORMATION OF THE WHU_HI_LONGKOU DATASET

No	Class	Train	Val	Test
1	Corn	35	35	34441
2	Cotton	9	9	8356
3	Sesame	4	4	3023
4	Broad-leaf soybean	64	64	63084
5	Narrow-leaf soybean	5	5	4141
6	Rice	12	12	11830
7	Water	68	68	66920
8	Roads and houses	8	8	7108
9	Mixed weed	6	6	5217
	Total	211	211	204120

TABLE VI
EACH CATEGORY INFORMATION OF THE BOTSWANA DATASET

No	Class	Train	Val	Test
1	Water	3	3	264
2	Hippo grass	2	2	97
3	Floodplain grasses 1	3	3	245
4	Floodplain grasses 2	3	3	209
5	Reeds	3	3	263
6	Riparian	3	3	263
7	Firescar	3	3	253
8	Island interior	3	3	197
9	Acacia woodlands	4	4	306
10	Acacia shrublands	3	3	242
11	Acacia grasslands	4	4	297
12	Short mopane	2	2	177
13	Mixed mopane	3	3	262
14	Chalcedony	1	1	93
	Total	40	40	3168

total of 3248 effective pixels. The dataset contains 14 object categories representing the land cover types of seasonal swamps, occasional swamps, and dry woodlands located at the far end of the delta. The details and division of each category of Botswana are shown in Table VI.

B. Experimental Settings

Several state-of-the-art classification methods utilized in HSIs are compared with the proposed GPF to demonstrate its superiority, including support vector machine (SVM), which is a machine learning-based method; CNN-based methods, i.e., A^2S^2KResNet [17], SSRN [16], and exploring 3-D–2-D CNN feature hierarchy (HybridSN) [15]; GNN-based methods, i.e., GCN [30]; and the methods of combining CNN and GNN, i.e., CNN-enhanced GCN (CEGCN) [51] and weighted feature fusion of convolutional neural network and GAT (WFCG) [19]. To ensure the optimal performance of the aforementioned methods, all experiments are conducted under the same environmental conditions, and the hyperparameters are set in accordance with those specified in this article.

During the experiments, the five datasets are divided into a training set, validation set, and test set; the details are shown in Tables II–VI. Among the five datasets, the Indian Pines and Botswana datasets have fewer labels, and Indian Pines also has extremely unbalanced labels between different categories, i.e., the number of pixels in Soybean-mintill is more than 100 times that of Oats (2455 versus 20), which causes great difficulties in classification. In order to ensure sufficient training data, only 1% of the labeled data are utilized as the training set for the Indian Pines and Botswana datasets. The validation set is comprised of 1% of the labeled data, while the remaining 98% of the labeled data are used for the test set.

For PaviaU, Salinas, and WHU_Hi_LongKou, each category has a large number of labels. If the split method is consistent with Indian Pines and Botswana, multiple models will yield too high a classification accuracy to compare their performance. Therefore, to ensure adequate labeled data for all categories, this article utilizes rounding calculations to distribute the data and set 0.1% of the label volume as the training and validation sets, with the remaining data reserved as the test set for the three datasets. All methods on each dataset use the same data for training, validation, and testing to allow fair comparison between them.

In the proposed model, the segmentation scale is fixed to 100, the learning rate is set to $2e-3$, and the number of training epochs is set to 300. *l2-norm* regularization of $1e-4$ is set in order to stabilize the network training process and suppress overfitting. In FGNN, hidden layer neurons are set to 60, the number of GAT heads is set to 8, and the output dimension is set to $[HW, 64]$, where HW is the total number of pixels. In PNN, the band selection rate is set to 0.98, and the kernel size of the band selection module is 1×1. The kernel size of the first point-depth Convolution layer is set to 3×3, and the second is set to 5×5. The output dimension of the PNN is also set to $[HW, 64]$. Finally, the fusion rate ω of the two branches is set to 0.2.

In order to systematically and objectively evaluate the classification performance of the proposed GPF, this article uses four evaluation indicators: each-category accuracy, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). A visual analysis of the experimental results is provided for each dataset. All experiments are averages of eight runs. We implement the algorithm in Python 3.8.0 and PyTorch 1.7.0. The hardware used in the experiment is i7-8700K CPU and NVIDIA A100 Tensor Core GPU.

C. Experimental Results

In this section, we discuss the experimental results of different methods on five datasets. The details of classification accuracy are given in Tables VII–XI. The classification maps obtained by these methods are shown in Figs. 6–10. The results highlighted in boldface represent the optimal outcomes.

1) Indian Pines Dataset: The experimental results on the Indian Pines dataset are shown in Table VII, and Fig. 6 shows the classification diagrams of various methods. The GPF achieves the highest classification accuracy. Due to the

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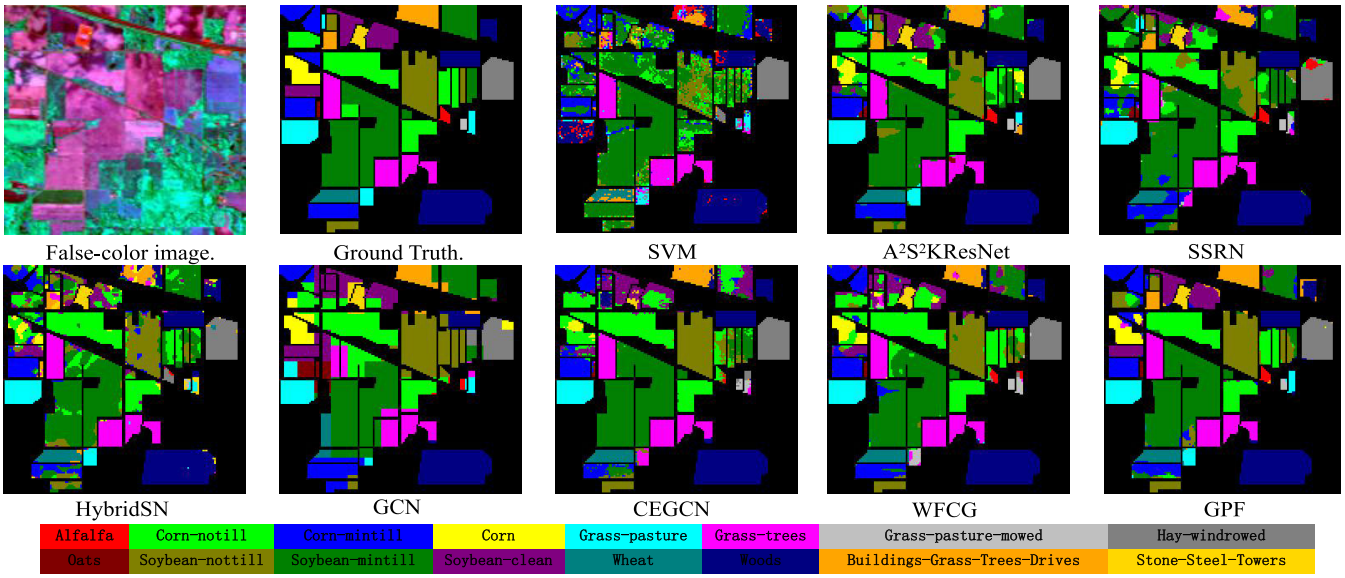


Fig. 6. False-color image, ground truth, and classification maps obtained by different methods for the Indian Pines dataset.

TABLE VII

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE INDIAN PINES DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	SVM	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
class 1	77.59 ± 12.01	90.57 ± 0.46	72.91 ± 0.94	43.75 ± 8.84	28.64 ± 4.11	24.24 ± 8.57	90.00 ± 3.16	**91.14 ± 4.96**
class 2	43.33 ± 6.40	80.18 ± 3.79	78.93 ± 2.27	71.27 ± 3.50	80.43 ± 4.63	85.43 ± 2.31	84.09 ± 3.96	**92.14 ± 2.45**
class 3	41.17 ± 3.97	**90.58 ± 1.01**	63.00 ± 2.98	71.37 ± 6.47	86.42 ± 1.54	80.75 ± 5.53	85.53 ± 10.13	83.47 ± 4.27
class 4	53.10 ± 18.37	82.98 ± 3.97	84.22 ± 2.70	56.24 ± 5.55	84.81 ± 16.81	33.33 ± 21.06	86.84 ± 4.49	**93.55 ± 6.34**
class 5	58.31 ± 1.04	95.38 ± 5.27	81.22 ± 1.17	**97.12 ± 1.26**	65.33 ± 0.78	65.33 ± 0.00	85.48 ± 8.41	93.43 ± 1.46
class 6	82.37 ± 5.73	96.51 ± 3.15	90.41 ± 3.47	95.53 ± 2.95	97.10 ± 0.75	**97.48 ± 0.11**	95.24 ± 3.43	95.84 ± 2.20
class 7	93.46 ± 1.81	75.88 ± 4.22	60.77 ± 5.30	83.66 ± 11.61	**100.00 ± 0.00**	75.64 ± 10.1	93.46 ± 2.36	99.23 ± 0.36
class 8	80.19 ± 4.12	98.73 ± 1.01	98.68 ± 0.59	92.09 ± 1.64	99.83 ± 0.02	**99.93 ± 0.01**	99.92 ± 0.06	99.83 ± 0.03
class 9	88.53 ± 4.81	52.40 ± 3.71	34.60 ± 0.63	17.76 ± 18.79	66.67 ± 0.33	51.85 ± 13.86	94.45 ± 2.84	**98.89 ± 0.56**
class 10	47.85 ± 3.15	82.28 ± 3.31	79.93 ± 2.09	64.53 ± 7.66	81.37 ± 4.99	**87.82 ± 1.29**	83.73 ± 7.21	83.50 ± 2.69
class 11	35.61 ± 8.83	81.27 ± 1.02	71.70 ± 0.87	81.02 ± 3.73	82.09 ± 3.11	84.59 ± 3.46	90.45 ± 3.89	**96.66 ± 1.15**
class 12	39.61 ± 4.11	**90.63 ± 0.26**	68.88 ± 4.99	69.15 ± 4.19	90.40 ± 9.89	53.76 ± 13.27	79.55 ± 8.73	90.17 ± 4.96
class 13	87.47 ± 4.43	98.45 ± 1.10	92.89 ± 3.91	89.56 ± 4.86	95.78 ± 1.78	**99.83 ± 0.04**	98.74 ± 0.56	98.09 ± 1.23
class 14	81.62 ± 1.55	94.49 ± 0.43	93.98 ± 1.53	89.68 ± 2.93	98.67 ± 1.05	**99.97 ± 0.03**	98.16 ± 1.01	99.63 ± 0.05
class 15	34.20 ± 7.23	**93.15 ± 1.12**	88.45 ± 1.39	80.41 ± 0.66	76.72 ± 2.89	74.60 ± 2.97	86.56 ± 5.04	76.22 ± 7.67
class 16	**92.56 ± 0.43**	91.30 ± 2.53	88.51 ± 0.16	75.29 ± 4.56	71.43 ± 0.65	54.95 ± 6.33	79.23 ± 6.32	85.82 ± 8.61
OA(%)	52.89 ± 1.96	87.14 ± 0.15	78.28 ± 0.39	79.15 ± 2.48	85.59 ± 1.41	83.67 ± 0.53	89.24 ± 1.19	**92.72 ± 1.07**
AA(%)	64.81 ± 2.25	87.17 ± 0.86	78.07 ± 0.78	73.65 ± 3.21	81.60 ± 3.16	73.09 ± 1.97	89.46 ± 3.26	**92.35 ± 1.39**
Kappa × 100	47.48 ± 1.98	85.24 ± 0.42	74.91 ± 0.41	76.15 ± 2.85	83.66 ± 1.59	81.34 ± 0.62	87.74 ± 1.34	**91.67 ± 1.23**

TABLE VIII

COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE PAVIAU DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	SVM	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
class 1	66.27 ± 2.28	70.22 ± 3.26	65.90 ± 3.45	77.33 ± 4.38	72.10 ± 4.27	**98.75 ± 0.93**	93.60 ± 2.69	95.37 ± 3.15
class 2	51.57 ± 4.05	91.48 ± 0.28	84.74 ± 3.17	94.76 ± 2.87	95.49 ± 1.31	**99.71 ± 0.11**	96.77 ± 2.31	99.63 ± 0.05
class 3	67.51 ± 3.08	79.84 ± 4.52	83.75 ± 0.89	66.80 ± 1.72	71.37 ± 3.59	**99.98 ± 0.02**	74.95 ± 3.87	83.86 ± 5.31
class 4	83.18 ± 4.68	**99.34 ± 0.60**	93.58 ± 0.58	96.89 ± 0.84	20.44 ± 3.00	80.14 ± 3.30	89.26 ± 2.03	87.42 ± 6.08
class 5	98.53 ± 0.90	98.53 ± 0.57	98.73 ± 0.26	97.68 ± 1.05	99.38 ± 1.03	99.93 ± 0.36	99.81 ± 0.59	**99.99 ± 0.02**
class 6	53.98 ± 2.62	84.53 ± 1.96	88.21 ± 1.21	81.23 ± 1.24	72.49 ± 3.12	96.25 ± 2.45	97.91 ± 1.53	**98.31 ± 1.09**
class 7	78.96 ± 3.06	97.37 ± 0.92	55.49 ± 0.28	66.52 ± 5.27	74.53 ± 4.20	18.07 ± 3.07	87.45 ± 4.24	**92.16 ± 7.42**
class 8	67.85 ± 2.91	60.92 ± 0.49	53.98 ± 0.33	53.30 ± 4.91	83.16 ± 3.06	83.03 ± 2.26	88.67 ± 3.11	**95.24 ± 3.87**
class 9	**99.75 ± 0.35**	97.42 ± 4.22	94.84 ± 1.61	51.71 ± 2.03	27.38 ± 2.87	71.54 ± 4.51	65.26 ± 5.08	81.98 ± 14.75
OA(%)	61.96 ± 2.87	83.98 ± 0.95	78.41 ± 0.79	83.69 ± 0.23	79.50 ± 1.58	93.18 ± 1.37	93.22 ± 1.76	**96.18 ± 0.67**
AA(%)	74.18 ± 2.26	86.63 ± 0.65	79.91 ± 0.61	76.24 ± 0.53	68.48 ± 1.67	83.05 ± 1.83	88.19 ± 2.28	**92.65 ± 1.32**
Kappa × 100	53.36 ± 3.03	78.26 ± 0.39	70.08 ± 0.88	78.30 ± 0.94	72.50 ± 1.81	90.86 ± 1.61	91.07 ± 1.98	**94.91 ± 0.91**

small samples in this dataset, the classification results of all methods are not ideal. It can be observed that SVM achieves the worst classification performance due to its own limitations, while the other two CNN-based models, SSRN and HybridSN, despite their strong expressive power, still suffer from small samples. Also, A^2S^2KResNet, due to the introduction of

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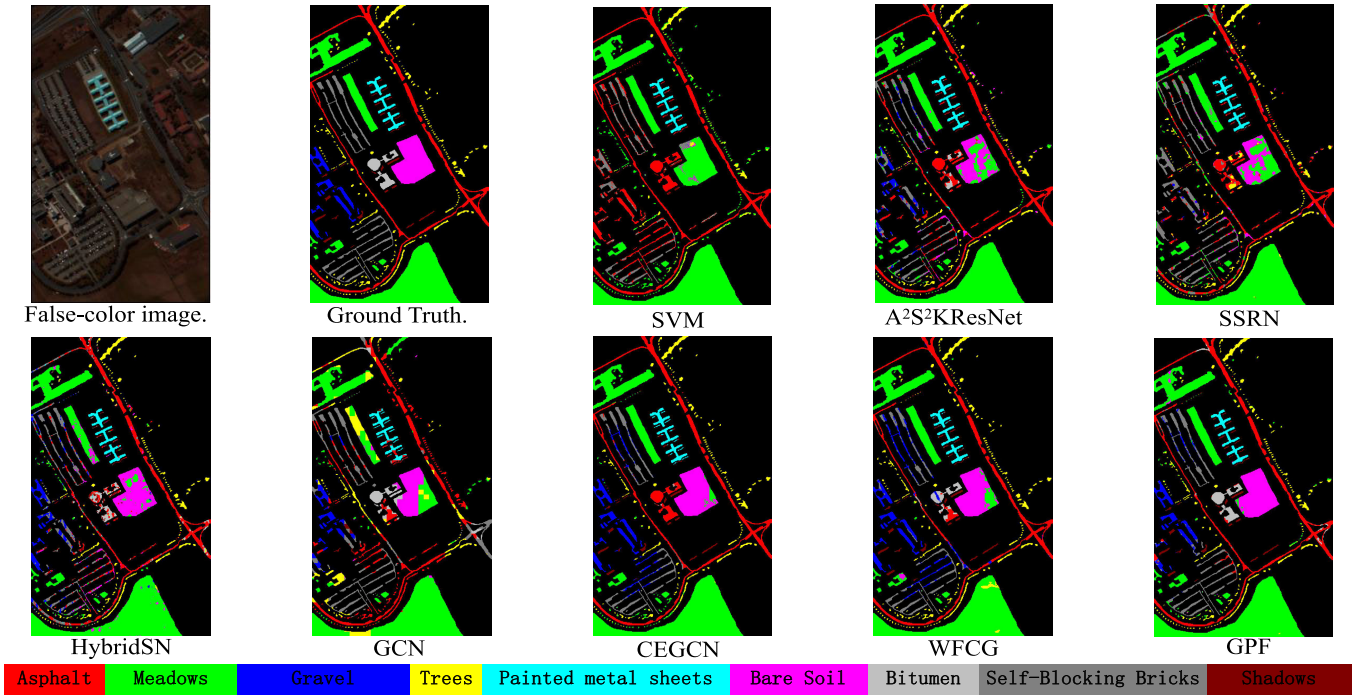


Fig. 7. False-color image, ground truth, and classification maps obtained by different methods for the PaviaU dataset.

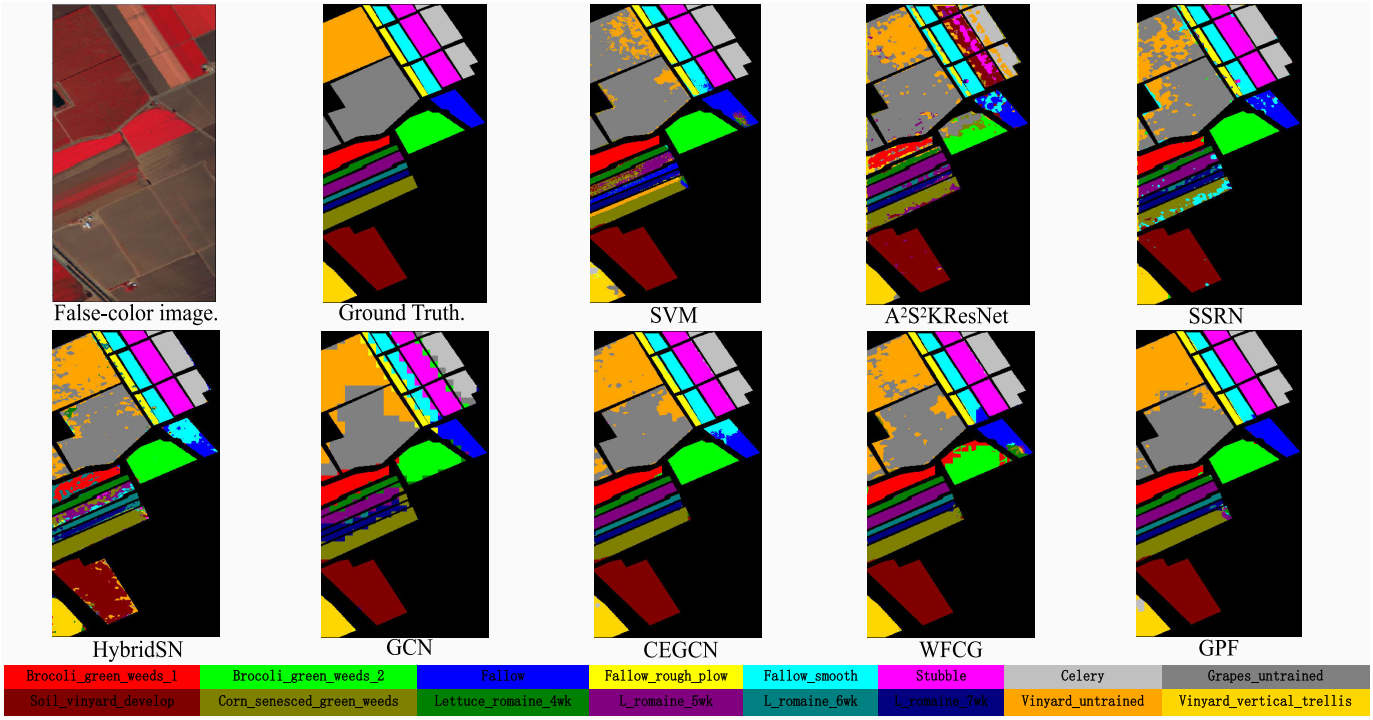


Fig. 8. False-color image, ground truth, and classification maps obtained by different methods for the Salinas dataset.

residual connections and attention mechanism, achieves a higher classification accuracy. The GNN-based method still achieves higher classification accuracy in the case of small samples. It is worth mentioning that the experimental effect of GCN on this dataset is better than that of CEGCN. CEGCN adopts the method of cascading feature fusion, which may cause certain error classifications with small samples. The classification performance of WFCG is relatively better; however, the classification map reveals that some regions, such as Soybean-clean, are not classified smoothly, which increases abnormal noise. In contrast, after superpixel composition, FGNN uses two operators to extract features to ensure classification accuracy in large target irregular regions and reduce classification errors caused by abnormal noise. In addition,

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TABLE IX
COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE SALINAS DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	SVM	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
class 1	96.27 ± 0.78	**99.98 ± 0.13**	99.24 ± 0.05	93.55 ± 3.33	94.21 ± 4.98	99.12 ± 0.74	98.42 ± 2.22	97.62 ± 1.74
class 2	95.11 ± 1.33	98.83 ± 0.36	98.15 ± 1.32	98.58 ± 1.02	90.48 ± 2.31	**100.00 ± 0.00**	99.94 ± 0.14	99.91 ± 0.12
class 3	76.39 ± 15.68	98.83 ± 1.01	**99.92 ± 0.04**	92.66 ± 4.34	94.07 ± 5.05	99.56 ± 0.31	72.47 ± 19.45	99.66 ± 0.95
class 4	98.82 ± 0.80	95.95 ± 2.39	91.53 ± 1.56	94.11 ± 2.79	35.76 ± 6.35	94.65 ± 3.32	96.68 ± 7.02	**99.95 ± 0.06**
class 5	89.11 ± 6.85	83.08 ± 4.56	81.87 ± 9.24	81.85 ± 8.56	88.77 ± 5.13	95.66 ± 2.97	**95.86 ± 4.56**	93.15 ± 1.29
class 6	98.08 ± 0.56	98.81 ± 0.35	97.21 ± 0.18	98.08 ± 0.92	84.43 ± 2.43	98.68 ± 0.37	98.66 ± 2.65	**99.89 ± 0.18**
class 7	98.61 ± 0.88	**100.00 ± 0.00**	99.31 ± 0.54	99.58 ± 0.18	77.18 ± 1.83	99.98 ± 0.03	99.34 ± 1.19	99.86 ± 0.15
class 8	53.51 ± 2.27	76.49 ± 5.26	76.07 ± 6.09	86.00 ± 4.07	82.06 ± 2.75	**92.92 ± 1.49**	87.32 ± 6.55	90.41 ± 2.84
class 9	95.41 ± 1.33	83.75 ± 7.89	90.67 ± 0.67	97.57 ± 1.03	98.51 ± 1.53	**99.99 ± 0.02**	**99.99 ± 0.03**	99.98 ± 0.06
class 10	76.78 ± 6.59	88.18 ± 2.86	95.82 ± 0.65	87.51 ± 5.39	91.41 ± 1.79	92.61 ± 1.77	92.45 ± 5.97	**93.80 ± 1.31**
class 11	94.82 ± 0.47	97.63 ± 0.55	97.56 ± 3.21	88.28 ± 5.23	85.15 ± 4.83	97.02 ± 1.85	98.66 ± 2.35	**99.13 ± 1.01**
class 12	99.78 ± 0.12	78.83 ± 7.89	77.48 ± 7.55	90.44 ± 4.78	84.82 ± 2.09	99.97 ± 0.05	99.75 ± 0.66	**100.00 ± 0.00**
class 13	98.53 ± 0.41	95.71 ± 1.02	82.61 ± 8.66	46.66 ± 4.85	36.32 ± 0.98	6.60 ± 4.21	96.70 ± 4.87	**99.08 ± 1.05**
class 14	87.97 ± 0.46	91.02 ± 3.56	91.57 ± 7.26	63.49 ± 8.81	91.09 ± 1.11	**97.87 ± 1.49**	96.87 ± 2.58	95.41 ± 2.76
class 15	69.91 ± 2.82	69.64 ± 9.65	76.47 ± 9.24	79.53 ± 5.87	85.80 ± 6.01	78.05 ± 12.23	86.88 ± 5.56	**91.61 ± 3.41**
class 16	87.60 ± 12.68	72.52 ± 2.66	90.39 ± 6.56	76.59 ± 8.02	**98.89 ± 2.32**	93.92 ± 3.55	94.21 ± 5.97	84.14 ± 6.65
OA(%)	81.41 ± 0.59	82.97 ± 1.45	84.96 ± 1.92	88.07 ± 0.53	85.60 ± 0.79	92.75 ± 2.42	93.29 ± 0.75	**95.38 ± 0.88**
AA(%)	88.55 ± 1.08	89.33 ± 0.65	90.37 ± 0.85	85.91 ± 0.69	82.43 ± 0.59	90.41 ± 1.35	94.64 ± 1.14	**96.48 ± 0.62**
Kappa × 100	79.43 ± 0.67	80.91 ± 0.59	83.16 ± 0.90	86.70 ± 0.92	83.99 ± 0.88	91.91 ± 2.72	92.54 ± 0.83	**94.86 ± 0.98**

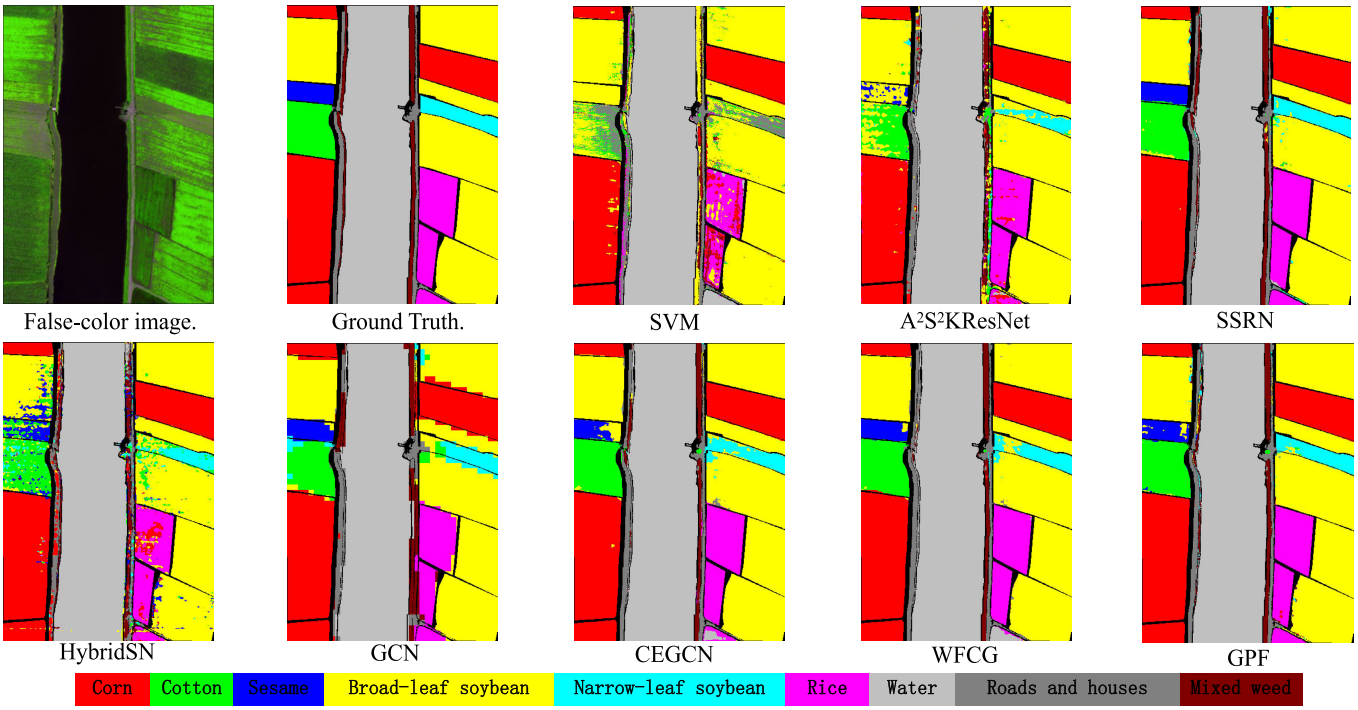


Fig. 9. False-color image, ground truth, and classification maps obtained by different methods for the WHU_Hi_LongKou dataset.

PNN performs feature extraction in small target regular regions and generates pixel-level features to supplement the shortcomings of FGNN in small target regions. Experimental results have shown that GPF is capable of generating a classification map that is more accurate and closer to the ground truth.

2) PaviaU Dataset: For the PaviaU dataset, it is worth mentioning that although the OA of SVM is the lowest, it has achieved the highest classification accuracy on Shadows, indicating that SVM still has advanced classification capabilities in the case of few samples. The three CNN-based methods have low classification accuracy, because they only use pixel-level features for classification and ignore the adverse effects of fine features on classification. On the other hand, GCN solely relies on superpixels to suppress the negative impact of fine features. However, it fails to consider the errors generated by superpixel segmentation in small regular regions, leading to poor overall performance. Due to the combination of pixel-level and superpixel-level classification features, both CEGCN and WFCG achieve more than 93% accuracy on OA. However, the classification accuracy of CEGCN in the Bitumen area is too low, causing AA to be lower than WFCG. The GPF combines the advantages of GNN and is fused with CNN at the same time, achieving the highest accuracy in the four categories,

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TABLE X
COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE WHU_HI_LONGKOU DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	SVM	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
class 1	89.41 ± 3.87	98.58 ± 0.84	99.15 ± 0.34	95.19 ± 0.97	97.55 ± 0.32	99.48 ± 0.11	**99.68 ± 0.47**	99.33 ± 0.29
class 2	73.62 ± 5.21	85.04 ± 4.80	77.69 ± 2.16	64.94 ± 3.23	96.87 ± 1.43	97.73 ± 1.01	94.62 ± 5.22	**99.06 ± 0.63**
class 3	80.42 ± 1.99	94.65 ± 1.49	90.28 ± 3.29	39.47 ± 9.54	66.31 ± 7.18	89.28 ± 2.01	**90.89 ± 7.47**	88.42 ± 2.81
class 4	84.63 ± 2.19	90.01 ± 1.34	95.95 ± 3.22	91.04 ± 2.64	95.86 ± 0.64	**99.45 ± 0.09**	99.13 ± 0.38	99.11 ± 0.28
class 5	88.96 ± 3.74	86.41 ± 1.48	86.03 ± 0.97	53.17 ± 6.62	85.89 ± 6.68	**97.78 ± 0.39**	81.15 ± 12.65	90.68 ± 2.96
class 6	95.25 ± 1.10	97.32 ± 1.99	95.61 ± 4.91	95.37 ± 0.94	94.15 ± 1.98	**99.75 ± 0.12**	95.88 ± 4.26	99.55 ± 0.47
class 7	99.87 ± 0.03	99.43 ± 0.25	99.74 ± 0.06	98.88 ± 1.48	99.42 ± 0.16	**99.99 ± 0.04**	99.96 ± 0.04	99.97 ± 0.03
class 8	84.63 ± 1.48	81.55 ± 1.35	82.67 ± 2.17	76.75 ± 3.86	70.62 ± 5.57	88.32 ± 6.98	94.26 ± 3.66	**95.93 ± 2.03**
class 9	72.76 ± 0.88	**91.37 ± 7.97**	84.49 ± 6.32	64.07 ± 6.92	47.24 ± 3.98	61.08 ± 11.53	86.18 ± 6.67	86.15 ± 2.93
OA(%)	90.33 ± 0.33	94.48 ± 0.03	95.44 ± 0.33	90.08 ± 2.23	94.49 ± 0.42	98.02 ± 0.07	98.13 ± 0.51	**98.68 ± 0.05**
AA(%)	85.51 ± 0.35	91.59 ± 0.35	90.17 ± 0.74	73.76 ± 2.32	83.77 ± 1.22	92.54 ± 0.58	93.53 ± 1.46	**95.35 ± 0.51**
Kappa × 100	87.48 ± 0.43	92.65 ± 0.77	94.00 ± 0.16	86.82 ± 1.88	92.76 ± 0.52	97.40 ± 0.11	97.54 ± 0.67	**98.26 ± 0.07**

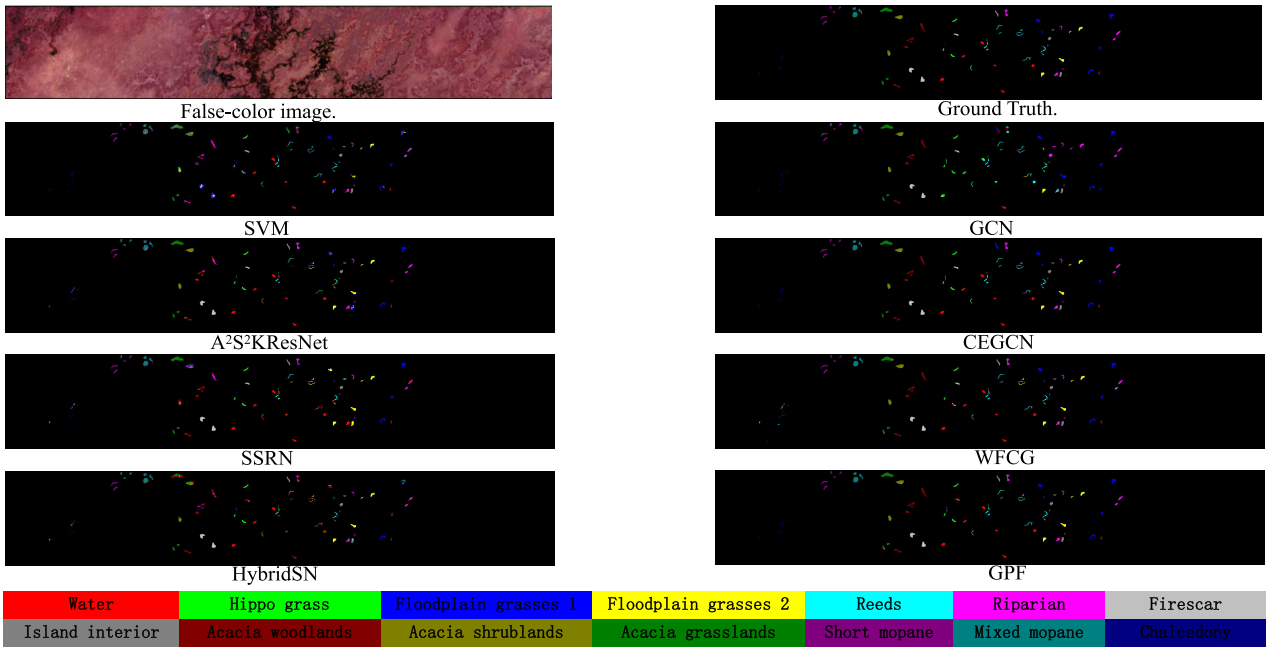


Fig. 10. False-color image, ground truth, and classification maps obtained by different methods for the Botswana dataset.

OA, AA, and Kappa. Also, compared with other methods, the classification map of GPF is less noisy and smoother.

3) Salinas Dataset: The Salinas dataset has the same 16 categories as the Indian Pines dataset. As shown in Table IX, SVM, A^2S^2KResNet, SSRN, and GCN perform poorly on this dataset, indicating the weaknesses of traditional machine learning methods and purely pixel- or superpixel-based methods when dealing with small samples. HybridSN shows a better classification effect on this dataset, but its AA is low, indicating that it fails to consider all regions leading to an unbalanced classification and the introduction of noise. The classification effect of CEGCN and WFCG is much higher than the above methods. But, the classification effect of CEGCN in the small sample area is poor, resulting in unbalanced classification, so AA is slightly lower. Compared with GPF, WFCG only uses the GAT operator and cannot extract features from multiple angles, resulting in lower classification accuracy than GPF. Through multiangle feature fusion, GPF can still achieve the best classification effect even when there is only one training sample in the Lettuce_romaine_6wk area.

4) WHU_Hi_LongKou Dataset: Table X demonstrates that, in this dataset, SVM surpasses HybridSN, indicating that traditional machine learning methods still possess competitiveness in some datasets. A^2S^2KResNet, which introduces the attention mechanism, achieves superior classification results compared with HybridSN. The attention mechanism compensates for the limitations of CNN by enabling it to focus on larger target regions. However, with the integration of residual connections, SSRN enhances its fitting ability and attains better OA value than A^2S^2KResNet and HybridSN. As a basic model, GCN attains comparable accuracy to that of A^2S^2KResNet. Moreover, it is evident that GNN displays stronger classification capabilities when confronted with a limited quantity of training samples. The other three models attain an accuracy exceeding 98%, yet due to the utilization of multiple feature extraction operators and the implementation

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TABLE XI
COMPARISON OF RESULTS BY DIFFERENT METHODS ON THE BOTSWANA DATASET, INCLUDING PER-CLASS ACCURACY, OA, AA, AND KAPPA

	SVM	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
class 1	97.37 ± 0.99	88.83 ± 0.71	69.27 ± 0.23	82.55 ± 0.12	41.89 ± 17.72	98.18 ± 2.03	99.09 ± 2.03	**99.81 ± 0.41**
class 2	56.80 ± 11.82	80.02 ± 0.12	84.02 ± 0.98	61.63 ± 0.16	91.34 ± 14.41	**100.00 ± 0.00**	76.29 ± 22.96	93.17 ± 9.02
class 3	92.74 ± 3.85	87.61 ± 0.13	83.55 ± 0.22	72.49 ± 0.08	79.59 ± 13.91	98.94 ± 1.95	**99.18 ± 1.42**	95.23 ± 9.05
class 4	75.01 ± 20.96	80.26 ± 0.12	75.39 ± 0.28	77.68 ± 0.11	30.15 ± 2.14	91.10 ± 12.53	**98.76 ± 2.78**	97.67 ± 3.75
class 5	74.56 ± 4.70	86.64 ± 0.11	73.20 ± 1.69	77.70 ± 0.12	82.97 ± 11.57	81.29 ± 4.66	82.28 ± 9.33	**96.72 ± 2.83**
class 6	28.57 ± 24.56	81.79 ± 0.12	62.05 ± 0.03	78.30 ± 0.13	45.17 ± 4.93	63.65 ± 18.22	88.06 ± 15.45	**94.01 ± 3.39**
class 7	74.80 ± 25.37	86.43 ± 0.08	91.66 ± 0.45	90.22 ± 0.60	91.46 ± 5.87	**100.00 ± 0.00**	99.68 ± 0.71	94.96 ± 5.93
class 8	83.50 ± 9.76	76.14 ± 0.14	75.75 ± 3.87	88.11 ± 0.09	56.36 ± 5.57	92.28 ± 7.97	84.77 ± 14.40	**100.00 ± 0.00**
class 9	52.52 ± 10.77	74.44 ± 0.08	77.88 ± 0.33	75.12 ± 0.13	99.48 ± 0.29	**100.00 ± 0.00**	99.48 ± 0.85	**100.00 ± 0.00**
class 10	39.19 ± 26.17	74.02 ± 0.25	89.29 ± 0.06	87.57 ± 0.07	99.84 ± 0.22	84.88 ± 20.71	98.35 ± 3.69	**100.00 ± 0.00**
class 11	70.64 ± 13.40	95.64 ± 0.02	98.81 ± 0.85	95.92 ± 0.47	**100.00 ± 0.00**	75.62 ± 1.05	89.63 ± 8.63	99.20 ± 1.53
class 12	90.26 ± 6.06	82.35 ± 0.12	88.28 ± 0.49	78.94 ± 1.15	80.68 ± 3.98	93.00 ± 8.91	**99.32 ± 0.93**	98.45 ± 2.89
class 13	54.59 ± 29.31	86.23 ± 0.08	80.73 ± 0.05	77.56 ± 1.03	95.72 ± 5.86	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
class 14	**98.91 ± 1.32**	88.31 ± 0.19	92.21 ± 0.72	86.87 ± 0.19	82.80 ± 0.32	82.80 ± 0.78	67.96 ± 23.74	86.02 ± 7.43
OA(%)	68.84 ± 3.18	79.20 ± 0.41	71.24 ± 0.04	78.78 ± 0.03	77.34 ± 2.49	89.79 ± 4.03	93.49 ± 1.63	**97.51 ± 1.61**
AA(%)	70.69 ± 3.22	83.48 ± 0.66	81.58 ± 0.02	80.76 ± 0.03	76.96 ± 2.41	90.12 ± 3.99	91.63 ± 2.65	**96.80 ± 1.92**
Kappa × 100	66.26 ± 3.48	77.48 ± 0.45	68.85 ± 0.05	77.04 ± 0.04	75.46 ± 2.69	88.94 ± 4.37	92.94 ± 1.76	**97.30 ± 1.75**

of PSA, GPF outperforms CEGCN and WFCG in terms of classification performance. However, CEGCN's insufficient fusion leads to notable discrepancies in the classification accuracy of diversified land features, resulting in a low AA value. As illustrated in Fig. 9, GPF exhibits lower classification errors and introduces less noise.

5) Botswana Dataset: As a highly challenging classification dataset, it can be seen from Table XI that many models encounter classification difficulties on the Botswana dataset. Due to the extremely low number of training samples, none of SVM, A^2S^2KResNet, SSRN, and HybridSN reach 80%, and it can be seen from Fig. 10 that these four models have large misclassifications. Although the attention mechanism is added and has played a certain role, due to the limitations of the attention mechanism, the classification performance of A^2S^2KResNet is restricted. After merging with CNN, the classification performance of CEGCN has been greatly improved compared with GCN, and all four types of ground objects are classified correctly. However, due to its insufficient fusion, the performance of CEGCN cannot be further improved. Under the influence of the fusion rate, WFCG has achieved improved and balanced classification results. However, relying solely on GAT for feature extraction from large target regions is irrational. In contrast, GPF achieves 97.51% accuracy due to its powerful feature extraction ability, and the classification map is also closer to the ground truth.

D. Influence of Residual Connection

Two residual connections are established in the PNN, as depicted in Fig. 11. To investigate the impact of residual connections, four different experiments were performed, with or without residual connections recorded as (0, 0), (0, 1), (1, 0), and (1, 1), where (1, 0) means that the residual connection is set at the first place, and not set at the second place. Each set is performed eight runs separately, and the experiment records are shown in Table XII.

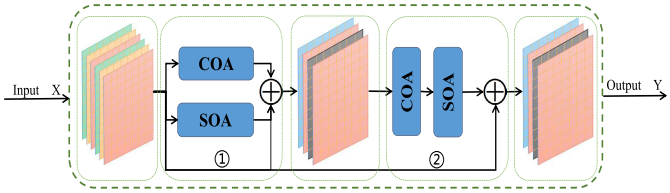


Fig. 11. Schematic map of residual connection.

TABLE XII
OA (%) OF THE FOUR SETTINGS ON EACH DATASET

	(0, 0)	(1, 0)	(0, 1)	(1, 1)
Indian Pines	78.65 ± 1.21	83.25 ± 0.44	86.10 ± 0.81	**92.72 ± 1.07**
PaviaU	87.99 ± 0.78	87.66 ± 0.24	93.05 ± 1.30	**96.18 ± 0.67**
Salinas	88.06 ± 0.45	90.88 ± 0.83	93.16 ± 1.61	**95.38 ± 0.88**
WHU_Hi_LongKou	97.10 ± 1.36	98.15 ± 2.35	98.25 ± 0.57	**98.68 ± 0.05**
Botswana	93.06 ± 2.39	78.79 ± 3.35	90.25 ± 1.56	**97.51 ± 1.61**

It can be seen from the table that (1, 1) has the best effect on all five datasets, which verifies the effect of residual connections. We find that the OA values for (0, 1) are better than those for (1, 0) and in four datasets are better than those for (0, 0), but not in the Botswana dataset. The OA values for (1, 0) are better than that for (0, 0) in the three datasets, but not in the PaviaU and Botswana datasets. Therefore, residual connections do not always have a positive regulatory effect. Instead, when applied to the deep layers of a network, they can better help the model adjust parameters and improve model fitting with the data. Consequently, blindly incorporating residual connections in models is not advisable, and careful consideration of network depth and data distribution is necessary.

E. Analysis of COA and SOA Modules Arranged in Series and Parallel

The PNN contains two PSA blocks arranged in series–parallel based on COA and SOA. There are four

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TABLE XIII
OA (%) OF THE FOUR ARRANGEMENTS ON EACH DATASET

	S-S	S-P	P-S	P-P
Indian Pines	89.15 ± 2.66	86.23 ± 1.38	**92.72 ± 1.07**	88.26 ± 0.87
PaviaU	92.35 ± 1.71	92.25 ± 1.89	**96.18 ± 0.67**	89.12 ± 2.56
Salinas	93.13 ± 0.91	92.20 ± 0.45	**95.38 ± 0.88**	94.00 ± 1.11
WHU_Hi_LongKou	98.53 ± 0.32	98.11 ± 1.29	**98.68 ± 0.05**	97.99 ± 1.21
Botswana	97.10 ± 0.39	95.08 ± 1.95	**97.51 ± 1.61**	89.33 ± 3.47

possible configurations: S–S, S–P, P–S, and P–P, where S represents series and P represents parallel. When connecting in series, inspired by CBAM [37], we put the COA module in front of the SOA. Eight runs were carried out for each cases to obtain the average value, and the final OA values are shown in Table XIII.

As Table XIII shows, the arrangement of P–S has the highest OA value of the five datasets, and the remaining three arrangements reflect uncertainty. Among them, the S–S arrangement is superior to the S–P arrangement and is better than the P–P arrangement in four datasets (except for Salinas), while the S–P and P–P methods have mixed performance. Through the P–S arrangement, in the P-PSA module, the model extracts the most original features of this dimension from the perspective of space and channel and then integrates them to ensure that the integrity of the features is more in line with the original distribution. In S-PSA, COA first integrates into the channel dimension to adjust the distribution of channel information, and then, SOA performs feature regulation in the spatial dimension. The P–S arrangement can greatly keep the original features of the data unchanged while taking into account the channel and spatial dimensions. As a result, sending the original feature distribution to the deeper layer of the network helps facilitate network training, leading to improved classification accuracy.

F. Attention Mechanism Comparison

Attention mechanisms can encode the input features according to certain rules without significantly increasing the number of parameters, thereby improving the performance of the model. In order to verify the performance of PSA, we compared the attention-free mechanism and embedded squeeze-and-excitation networks (SE-Nets) [36], efficient channel attention for deep convolutional neural networks (ECA-Nets) [52], spatial pyramid attention network (SPA-Net) [53], criss-cross attention networks (CC-Nets) [54], global context networks (GC-Nets) [35], PSA networks (PSA-Net) [31], and CBAM [37] changes in model accuracy. Also, the corresponding results are shown in Fig. 12.

With embedding different attention mechanisms, the GPF still performs well. Through experiments, it can be found that PSA works best on five datasets, so embedding it into the GPF is selected as the preferred approach. Different from the original paper, the COA and SOA modules are arranged in parallel and then in series, which will further improve the ability of the model to extract complex features. In the parallel arrangement, the model independently extracts the most fundamental features from both the spatial and channel dimensions, ensuring the comprehensiveness and consistency of the feature information. This is followed by a serial arrangement where the COA module integrates the channel dimension to regulate the distribution of channel information, while the SOA module further adjusts the features in the spatial dimension based on this. To further optimize the feature extraction process, a residual connection is introduced to integrate the low-order features with the high-order features, resulting in a more nuanced and comprehensive representation of the data. By combining these techniques, the model is able to capture both the spatial and channel information of the input data, thereby enhancing its performance in HSIs classification.

Furthermore, it is noteworthy that disparate attention mechanisms have varying performance levels when applied to classification tasks involving distinct datasets. In addition, the high time complexity of CC-Net makes it infeasible for operation on the Botswana dataset, resulting in only four outcomes being presented for this dataset. Even without an embedded attention mechanism (NULL), GPF shows impressive classification capabilities, outperforming certain models that do include attention mechanisms. Hence, it is imperative to exercise caution in the selection of attention mechanisms for specific tasks, as improper employment may engender unfavorable consequences.

G. Influence of Training Sample Ratio

In order to prove the performance of GPF under different ratios of training samples, we set different training sample ratios and use the methods mentioned in *C* of Section IV to conduct experiments on the five datasets. In particular, the training sample ratio is set from 0.01 to 0.06 for the Indian Pines and Botswana datasets, while it is set from 0.001 to 0.006 for the remaining three datasets. The line chart of the comparison results is as in Fig. 13.

Fig. 13 reveals that the OA values of all models exhibit growth with an augmented training sample ratio. SVM, due to its intrinsic limitations in feature extraction, is outperformed by other models, barring the instance wherein its performance is marginally superior to HybridSN at a ratio of 0.001 in the WHU_Hi_LongKou dataset. Despite being a rudimentary model, GCN's performance across the five datasets remains remarkable, although it is outshone by A^2S^2KResNet, SSRN, and HybridSN as the ratio of training samples increases. Overall, A^2S^2KResNet, which incorporates residual connections and attention mechanisms, outperforms SSRN and HybridSN. Nevertheless, the three CNN-based models pale in comparison with fusion models, such as CEGCN, WFCG, and GPF. CEGCN's subpar performance, attributable to insufficient fusion, places it behind WFCG in most cases. GPF achieves exceptional performance on five datasets, particularly under low training sample ratios. Our results suggest that using multiple feature extraction operators, incorporating an appropriate attention mechanism, and designing a suitable feature fusion strategy can enhance the accuracy and stability of HSIs classification.

H. Visual Analysis of Feature Fusion

The GPF proposed contains two important classifiers, namely, the FGNN classifier of the GNN branch and the PNN

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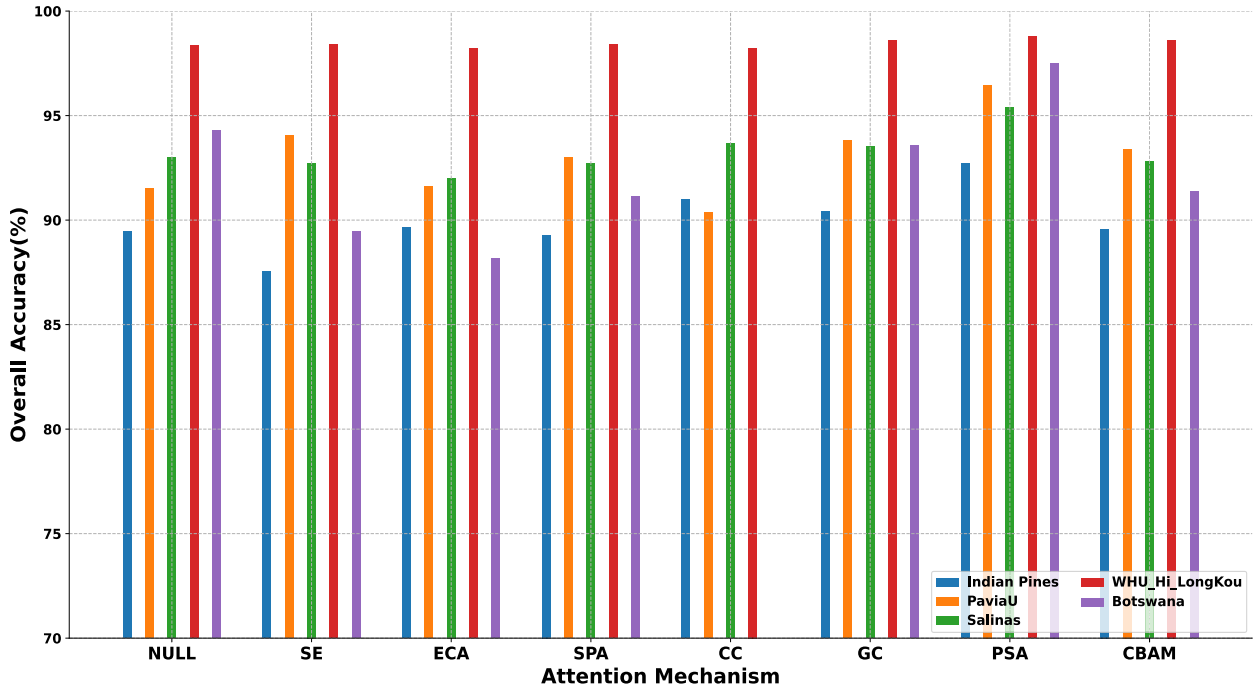


Fig. 12. Results of embedding with different attention modules on five datasets.

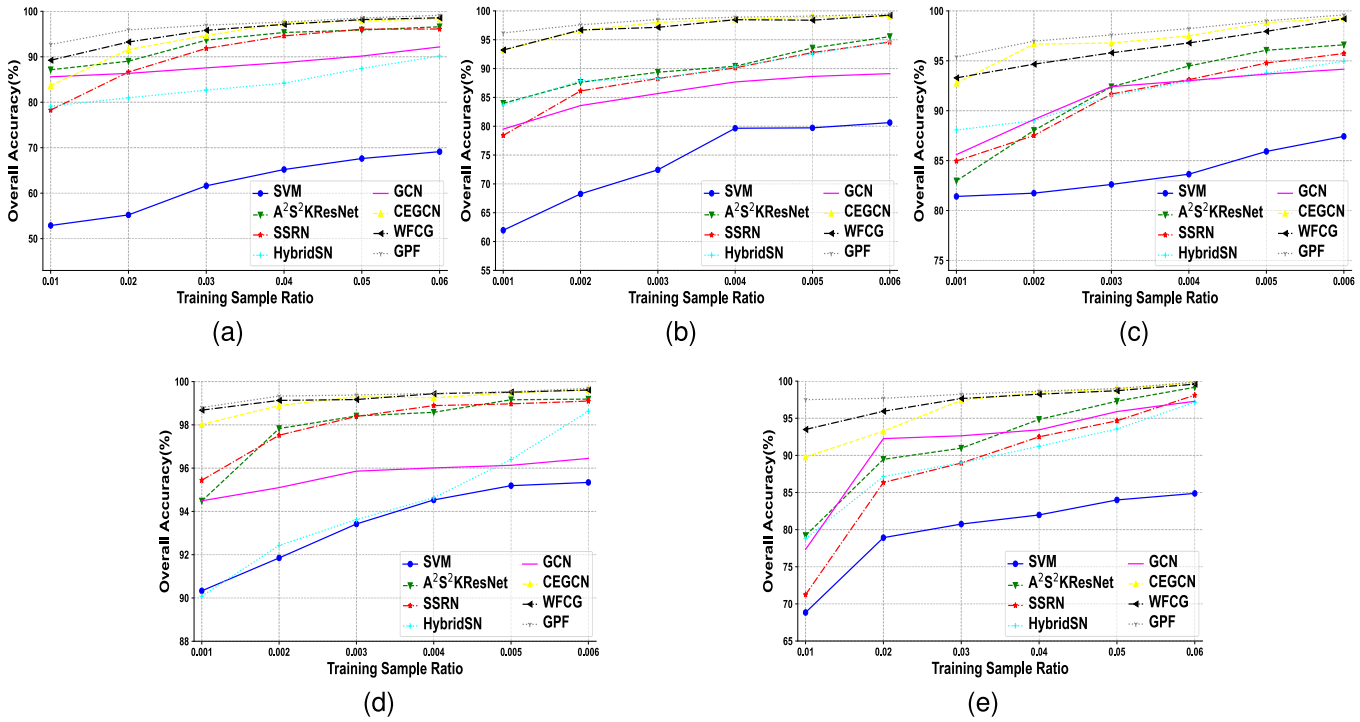


Fig. 13. Classification performance of each method with different training sample ratios on five datasets. (a) Indian Pines. (b) PaviaU. (c) Salinas. (d) WHU_Hi_LongKou. (e) Botswana.

classifier of the CNN branch. In order to study the properties of the two classifiers, we have carried out independent experiments on them and visualized the classification results, and the experimental settings are consistent with GPF. Figs. 14–18 are the comparison of classification maps on the five datasets.

The FGNN classifier demonstrates evident processing of superpixel-level features, leading to the emergence of jagged regions in the resultant classification map. This indicates excessive smoothness in the classification process, which may result in a large number of misclassifications in small target regions that have regular shapes. In contrast, the PNN operates at the pixel level and can precisely classify even small regions, generating more balanced classifications. However, this precision comes at the cost of introducing salt and pepper

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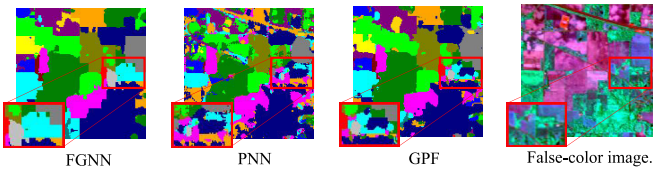


Fig. 14. Comparison of classification maps on the Indian Pines dataset.

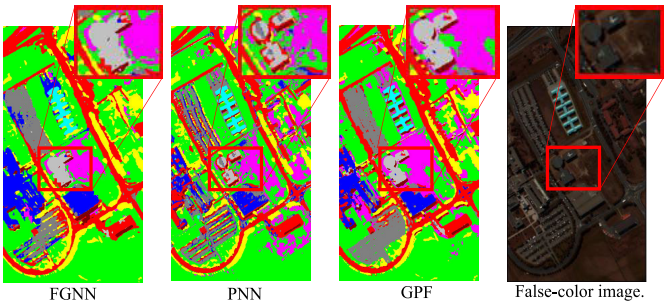


Fig. 15. Comparison of classification maps on the PaviaU dataset.

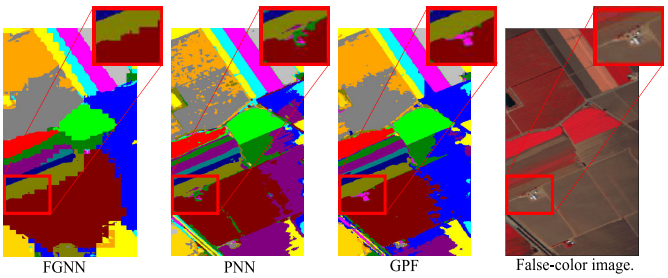


Fig. 16. Comparison of classification maps on the Salinas dataset.

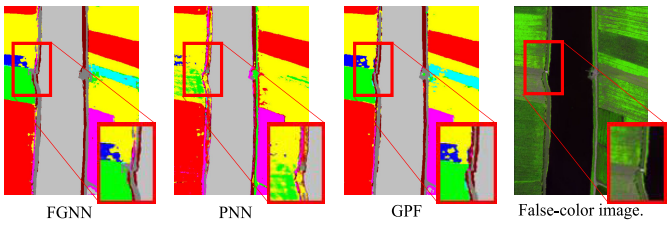


Fig. 17. Comparison of classification maps on the WHU_Hi_LongKou dataset.

noise, which may degrade the quality of the classification map.

To mitigate these issues, GPF integrates features from both FGNN and PNN to generate a classification map that closely approximates the ground truth. GPF utilizes the smoothing capabilities of FGNN for larger objects, while relying more on the refinement properties of PNN for smaller objects. As a result, GPF effectively balances the trade-off between precision and smoothness, while minimizing the impact of salt and pepper noise.

I. Influence of Segmentation Scale and Fusion Rate

The performance of GPF is influenced by both the segmentation scale and the fusion rate. A smaller segmentation scale can result in more noise due to the increased number of graph nodes, whereas an excessively large segmentation

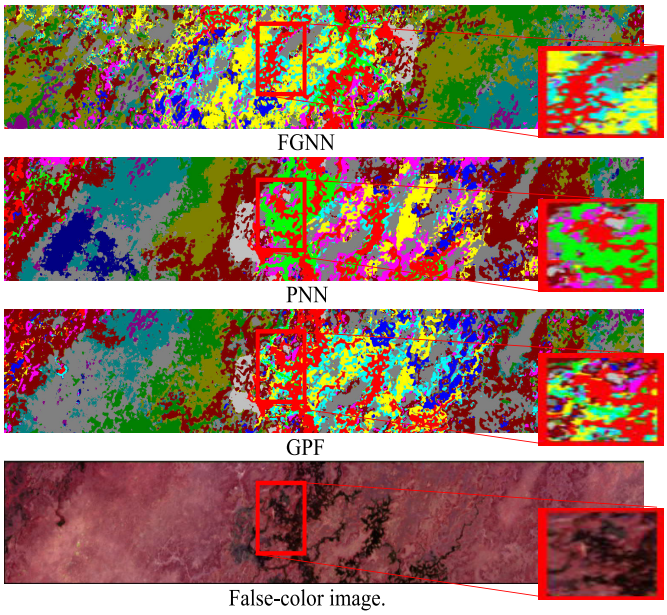


Fig. 18. Comparison of classification maps on the Botswana dataset.

scale can change the original data distribution of HSIs and make classification more difficult. The fusion rate determines the weighting of the two branches. Improper distribution of the fusion rate amplifies the limitations of both branches, contrary to the intended effect. Careful attention to selecting an appropriate segmentation scale and fusion rate is essential to optimize the performance of GPF.

To investigate the influence of segmentation scale and fusion rate, we conduct experiments on five datasets by setting the value of the segmentation scale to [100, 200, 300, 400, 500] and the value of fusion rate to [0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5], resulting in a total of 40 combinations. The final results are depicted in Fig. 19, showcasing the maximum OA value on each dataset through a blue dot.

As illustrated in Fig. 19, all five datasets demonstrate a phenomenon of high in the middle and low around, implying that the segmentation scale and fusion rate of polarization can have a negative impact. It is worth noting that the [100, 0.2] configuration achieves the highest OA value on the PaviaU, Salinas, and Botswana datasets, suggesting that these datasets are better suited for small-scale segmentation. However, the combination that yields the highest value on the other two datasets varies. In all experiments, the segmentation scale is set to 100, and the fusion rate is set to 0.2 to prevent overfitting.

J. Comparison on Other Metrics

In order to comprehensively assess the proposed GPF, it is inadequate to solely rely on OA, AA, and Kappa. Therefore, we evaluate the methods (excluding SVM) based on additional metrics, such as the number of parameters, floating-point operations (FLOPs), training time, testing time, and Z values of McNemar's test on all five datasets. Among them, when calculating the number of parameters and FLOPs, the three CNN-based models are calculated by batch size 1, while the four GNN-based models are trained by putting the whole

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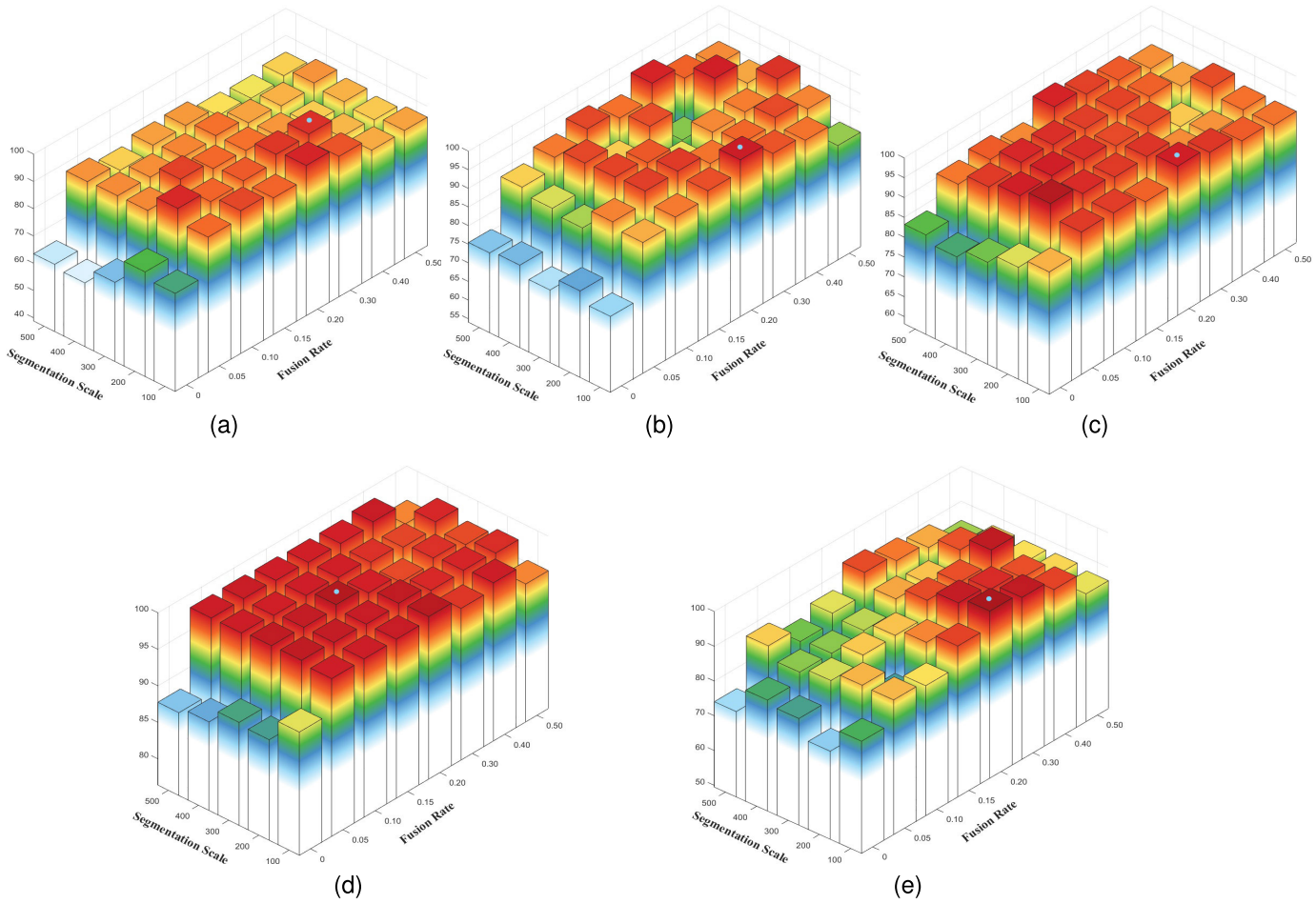


Fig. 19. Classification performance with different combinations of segmentation scale and fusion rate on five datasets. (a) Indian Pines. (b) PaviaU. (c) Salinas. (d) WHU_Hi_LongKou. (e) Botswana.

graph into the network. The experimental results for these metrics, conducted under the original training sample ratio, are presented in Table XIV. – signifies that GPF does not undergo McNemar's test with itself.

Through the parameter values, we can find that HybridSN contains the most parameters, because it stacks multiple 3-D convolutional layers for feature extraction, which adds a lot of FLOPs and redundant parameters in the process. The parameters and FLOPs of A^2S^2KResNet and SSRN are basically the same, but the training time and test time of A^2S^2KResNet are higher than SSRN and HybridSN. However, the duration of the running time is determined by multiple factors. It is constrained by the number of parameters, but it also depends on other hyperparameters, such as the batch size and the number of epochs. GCN is a basic GNN model, and its four indicators are kept at a low level. As fusion models, CEGCN, WFCG, and our GPF are comparable to the above three CNN-based models in terms of the number of parameters, but the FLOPs are much higher than the three models. This phenomenon occurs due to the addition of numerous operations between large matrices, which are present after constructing HSIs as graphs. In addition, the high resolution of HSIs results in a dramatic increase in graph nodes. Furthermore, CNN-based models predominantly increase the FLOPs in the channel dimension due to their fixed convolutional kernel size. In contrast, fusion models, such as GPF, primarily perform FLOPs in the spatial dimension. It is widely acknowledged that the spatial dimension of HSIs is substantially larger than their channel dimension, leading to a significant increase in FLOPs. Through the test time, we can find that the test time of the fusion models, such as GPF, is much lower than that of the CNN-based models, such as A^2S^2KResNet. During training, CNN-based models employ batch training, where a portion of the patches generated by HSIs are used for training, while the remaining are reserved for testing. In contrast, fusion models train the entire graph, with the only difference between the training and testing phases being the mask used for calculating the loss. As a result, the testing time for fusion models is equivalent to the training time for one epoch, which is significantly lower than that of CNN-based models. The number of parameters and FLOPs of GPF are higher than those of CEGCN and WFCG. This is due to GPF's use of both Res_GCN and GAT operators, which inevitably results in increased computational costs and longer runtime. However, this design choice also leads to higher classification performance for GPF.

The Z values presented in Table XIV represent the McNemar's test results comparing six models with the GPF.

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TABLE XIV
COMPARISON OF COMPUTATIONAL COST, RUNNING TIME, AND Z VALUES OF COMPETITORS ON FIVE DATASETS

Dataset	Metrics	A^2S^2KResNet	SSRN	HybridSN	GCN	CEGCN	WFCG	GPF
Indian Pines	P (M)	0.37	0.36	3.55	0.14	0.17	0.17	0.29
	F (GMac)	0.17	0.16	0.76	0.03	1.59	1.53	5.31
	Train time (s)	109.13	31.13	21.27	10.03	10.16	11.32	17.23
	Test time (s)	22.68	10.95	9.91	0.12	0.15	0.16	0.23
	Z value	16.03	35.15	33.31	17.47	23.45	4.83	–
PaviaU	P (M)	0.22	0.22	1.76	0.12	0.15	0.09	0.08
	F (GMac)	0.10	0.08	0.23	0.21	13.05	12.41	15.03
	Train time (s)	63.27	11.77	9.78	24.29	25.01	103.56	245.06
	Test time (s)	109.92	46.81	14.14	0.23	0.29	0.73	0.82
	Z value	45.21	56.59	41.87	53.81	22.93	21.69	–
Salinas	P (M)	0.37	0.37	3.62	0.14	0.17	0.10	0.30
	F (GMac)	0.17	0.16	0.36	0.11	8.45	8.15	28.83
	Train time (s)	94.85	20.28	14.38	16.51	17.23	40.26	69.56
	Test time (s)	325.53	185.62	54.37	0.17	0.19	0.41	0.57
	Z value	59.08	53.60	41.27	50.87	32.78	23.42	–
WHU_Hi_LongKou	P (M)	0.48	0.47	4.84	0.15	0.17	0.11	0.52
	F (GMac)	0.23	0.21	0.28	0.23	18.52	17.94	98.42
	Train time (s)	108.71	65.55	36.26	26.13	26.54	126.49	329.69
	Test time (s)	875.63	632.19	147.10	0.37	0.37	0.55	0.94
	Z value	45.69	38.37	55.92	43.45	23.62	19.56	–
Botswana	P (M)	0.29	0.28	2.54	0.12	0.16	0.09	0.16
	F (GMac)	0.14	0.11	0.20	0.34	25.68	24.83	51.98
	Train time (s)	37.46	15.81	10.47	47.68	55.62	325.77	409.11
	Test time (s)	7.23	3.27	1.95	0.44	0.46	0.58	0.69
	Z value	19.64	28.07	23.16	25.31	15.91	7.59	–

If $|Z| > 1.96$, it indicates a statistically significant difference between the two classifiers. Moreover, if Z is positive, it signifies that the GPF outperforms the compared model. As observed from Table XIV, all Z values are positive and greater than 1.96, indicating that the GPF outperforms the compared models. In addition, it is noteworthy that the Z value for WFCG is the smallest, suggesting that the performance of WFCG is closest to GPF.

V. CONCLUSION

In this article, a new model called GPF is proposed for the HSIs classification. The GPF consists of two important branches. In the GNN branch, we design a classifier named FGNN, which processes superpixel-level features acting on large irregular target regions. In contrast to prior research, the FGNN classifier incorporates two different feature extraction operators, namely, Res_GCN and GAT, to extract features from multiple angles. Moreover, it incorporates min, max, and weighted add feature aggregators to conduct multidimensional feature aggregation. Finally, 2-D convolutional layers are used for node updates. The other branch is the CNN branch, which is based on the PNN classifier. The PNN classifier is constructed based on point-depth convolution layers, which embed PSA blocks to capture pixel-level features in small, regular target regions. Residual connections have been integrated into the architecture to enhance the model's information processing capacity. The outputs from the two branches are fused using a weight-based approach, and the final classification results are obtained through the classification network. The GPF is evaluated on five well-known real datasets using several metrics, such as each-category accuracy, OA, AA, Kappa, computational cost, running time, and Z values of McNemar's test, and is compared with eight representative related methods. The experimental results show that GPF has comparable performance to the related methods.

In future work, the embedding of PSA block into GNN and the dynamic extraction of multiscale spatial information using a single-branch structure will be considered, so as to improve the classification accuracy and computational efficiency of the model while reducing the number of parameters.

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